RV College of Engineering®

Unit-II
ADVANCED ENERGY SYSTEMS

Go, change the world®

Outline

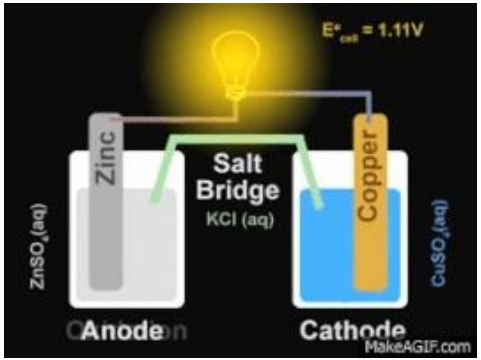
- This unit mainly focuses on the advanced energy systems used in day-today life.
- Basics of battery technology, commercial lithium battery used in electronic gadgets along with advanced lithium air battery.
- Innovation in the field of super capacitor and their importance in energy storage. Finally, it covers the green energy systems such as fuel cell and solar energy.
- Overall, this unit helps the learners in exploring the knowledge of energy storage devices for various applications in modern engineering practices.

Outline

Blow-up Syllabus		
Sl No	Topic to be taught	Duration in hr
1	Battery: Introduction, types, characteristics	1
2	Components/materials, construction, working and applications of Lithium cobalt oxide battery along with advantages and limitations	1
3	Construction and working of Metal air batteries along with their advantages and limitations.	1
4	Super-capacitors: Introduction, types (EDLC, pseudo capacitors, hybrid capacitors), with examples.	1
5	Construction and working mechanism of EDLC, pseudo capacitors, hybrid capacitors along with applications.	1
6	Energy conversion devices: Introduction, characteristics, materials, working and applications of H_2-O_2 and	1
7	Construction, working and applications of direct methanol fuel cells	1
8	Construction and working of Quantum dot sensitized solar cells.	1
9	Materials used in solar energy conversion	Self-study

Eelectro Chemistry

- Electrochemistry deals with chemical reactions that produce electricity and the changes associated with the passage of electrical current through matter.
- The reactions involve electron transfer, and so they are oxidation-reduction (or redox) reactions.
- Applications:









Battery



Duracell batteries



9v battery

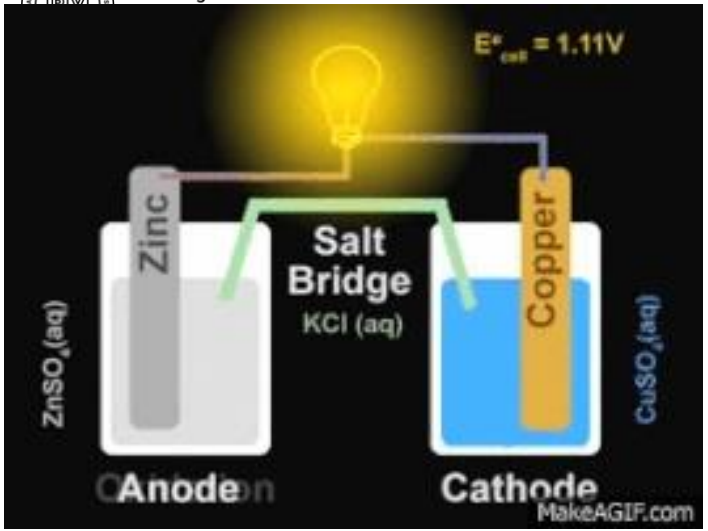


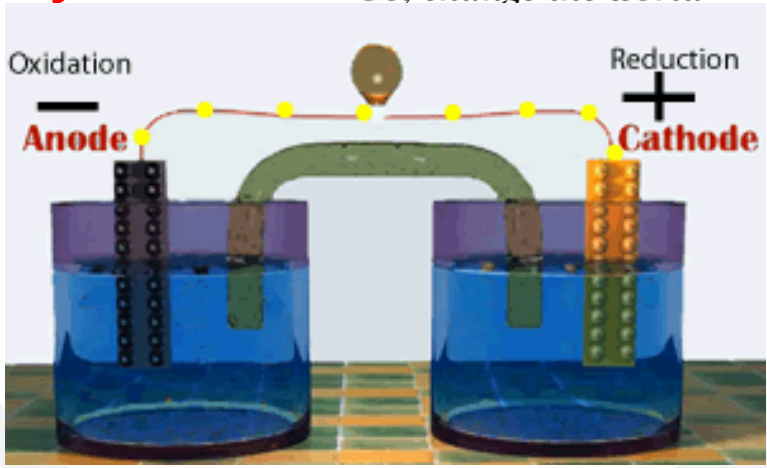
6v dry cell





Eelectrochemistry





Cell Reactions

$Zn^0_{(s)} \rightleftharpoons Zn^{2+}_{(aq)} + 2\,e^-$ +0.76 Oxidation

$Cu^{2+}_{(aq)} + 2\,e^- \rightleftharpoons Cu^0_{(s)}$ +0.34 Reduction

$Cu^{2+}_{(aq)} + Zn^0_{(s)} \rightleftharpoons Cu^0_{(s)} + Zn^{2+}_{(aq)}$ +1.10 Cell Reaction

Electrode: When any metal/gas in contact with its own ions form An electrode.

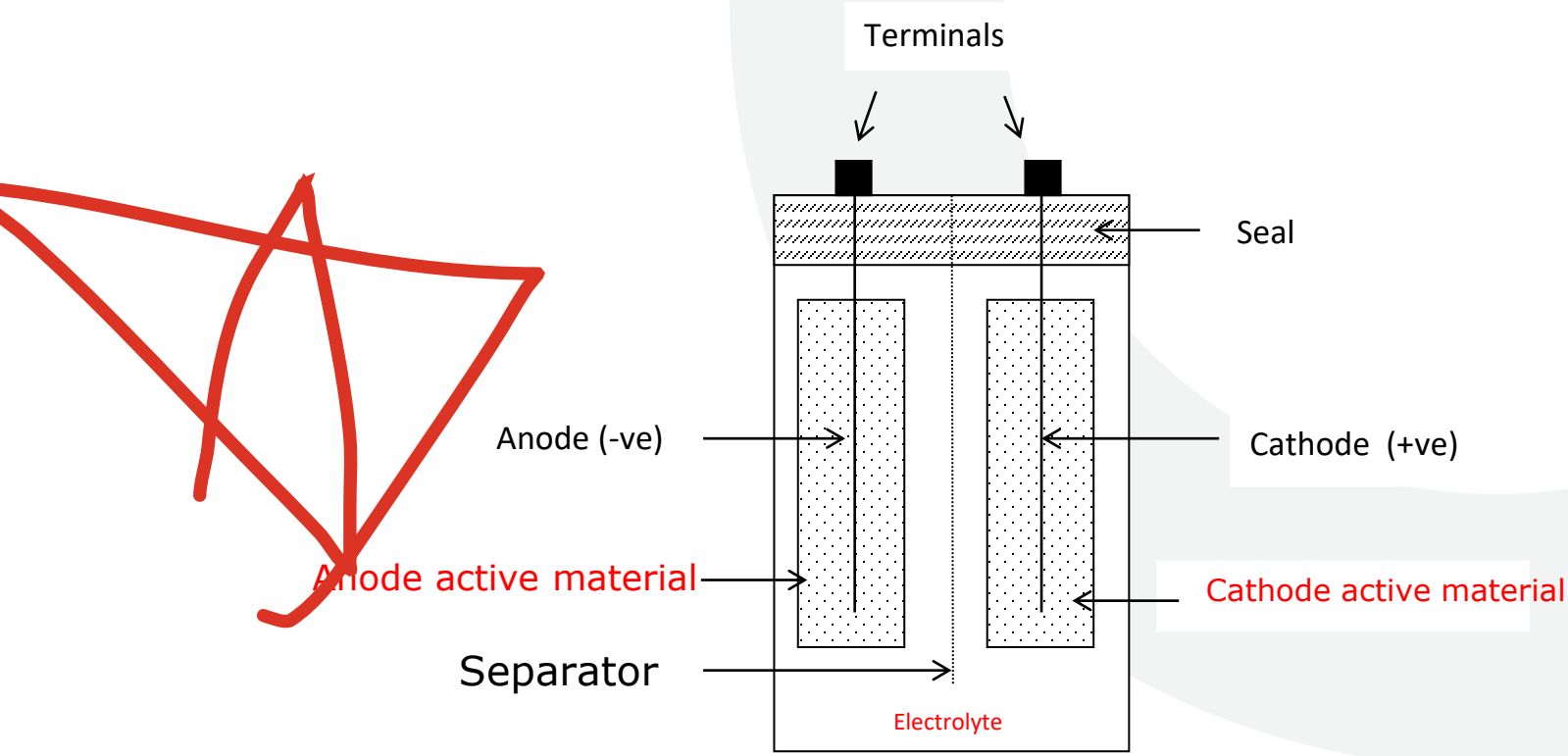
For Ex: Zn in contact with $ZnSO_4$

Cu in Contact with $CuSO_4$

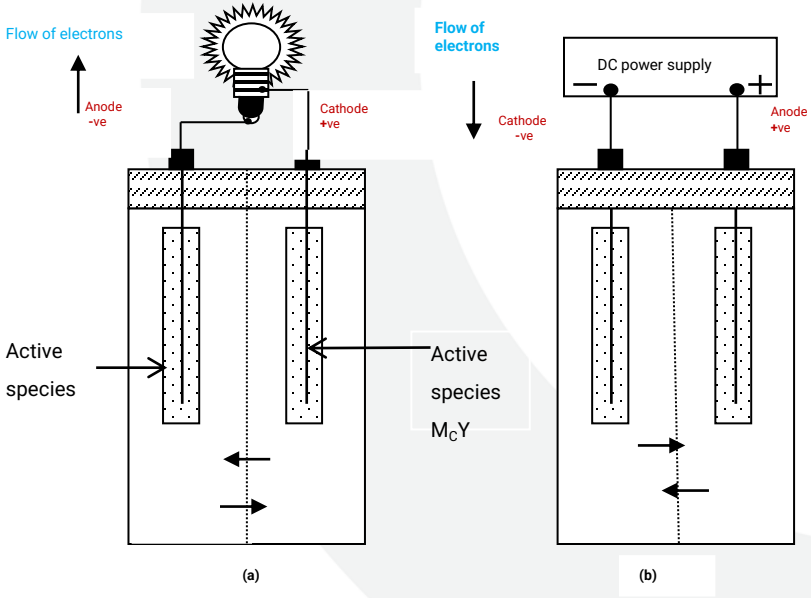
Cell: Combination of two electrodes

Battery: Two or more cells in combination.

COMPONENTS OF BATTERY



WORKING OF BATTERY



Electroactive material:

- When anode and cathode are connected by a metallic wire through load, spontaneous the reaction occurs at anode and cathode.
- Material which take part in the spontaneous reactions are called electroactive materials.

WORKING OF BATTERY

Discharging:

At anode

$M_a \longrightarrow M_a^{+n} + ne^-$

At Cathode

$M_cY + ne^- \longrightarrow M_c + Y^{-n}$

- During the reactions (Oxidation/Reduction), the electroactive materials are converted into products called electroinactive materials.
- The process in which electroactive materials are converted in to electroinactive materials is called discharging.
- During discharging the emf of a cell decreases.

WORKING OF BATTERY

Recharging (Charging)

At Cathode

$M_c + Y^{-n} \longrightarrow M_cY + ne^-$

At Anode

$M_a^{+n} + ne^- \longrightarrow M$

- In some cases, the electro inactive materials can be converted back to electroactive materials by passing electricity in opposite direction.
- This process is called charging of a battery.
- During charging the emf of a cell gradually increases and reaches its initial value.

CLASSIFICATION OF BATTERIES

Primary Batteries:

Electroinactive materials cannot be converted into electroactive materials, Irreversible batteries.

Ex: Dry cell, Alkaline Battery



Secondary Batteries:

Electroinactive materials can be converted into electroactive materials, reversible batteries by passing current in opposite direction.

Ex: Lead-acid battery, Ni-Cd Battery, Ni-MH Battery



Reserve Batteries :

These are inactive when not in use and may be activated in several ways prior to the application. In principle batteries without vital component.

Ex: Silver Chloride Cell, Mg/water/AgCl/Ag

Magnesium Battery, Mg/water/ Cu_2 Cl_2, Cu

CHARACTERISTICS OF BATTERY

1. Voltage (EMF)

According to Nernst equation at 298K

$$E = E^0 - \frac{2.303RT}{nF}\log Q$$

Where Q= [PRODUCT]/[REACTANT]

The emf of the battery depends on the following factors, E^o_{cell}, temperature and the concentration ratio given by Q.

(i) Where, $E^o_{cell} = E^0_R - E^0_L$

The difference in the standard electrode potentials of the two electrodes.

- As E^o_{cell} increase the e.m.f also increases for a given value of Q at constant temperature.
- In other words, the magnitude of e.m.f is large if the electrodes used in the cell are far apart in the electrochemical series.

(ii) From Nernst equation it is evident that as temperature T increases the EMF of a cell decreases.

(iii) Increase in the value of Q marginally decrease the e.m.f .

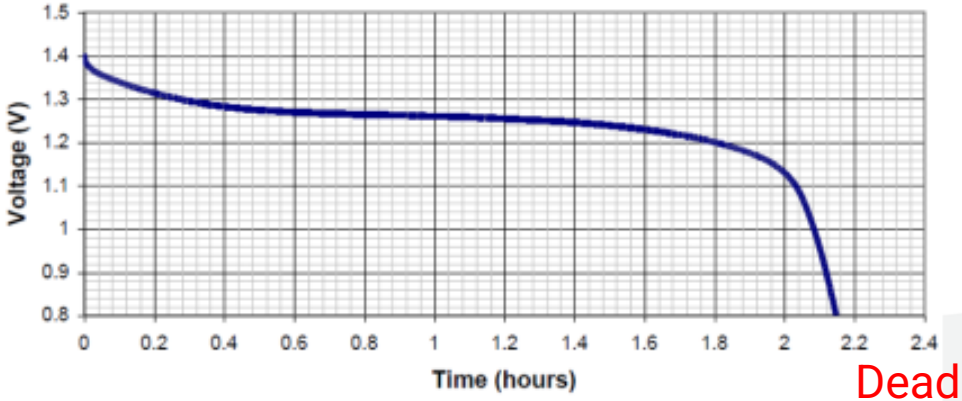
- This is because Q decreases when [products] decreases i.e when current is drawn from the cell.

CHARACTERISTICS OF BATTERY

1. Voltage (EMF)

Battery discharge curve





Dead state of battery

- The flat part of the curve is extendable.
- A good battery of the curve should have fairly long flat portion.

CHARACTERISTICS OF BATTERY

2. Current Capacity (C)

- The capacity is the charge that may be obtained from the battery.
- It is also defined as the amount of current generated per unit time.
- It is expressed in ampere hours [Ah] and it depends on the size of the battery.
- According to Faraday's law Capacity

$$C = wnF/M$$

Where, w = mass of the active material.

M = molecular mass of the active material.

3. Current

- Current is the rate at which the battery is discharging.
- For an efficient working of the cell there must be a large quantity of electro active species that help rapid electron transfer reaction.

CHARACTERISTICS OF BATTERY

4. Power Density:

- The power generated in a cell is given by P = iE,

where i is the current drawn from the cell and E is the cell voltage.

- To compare the power density of different cells a new factor power density is introduced.
- It is defined as the power generated per unit weight of the battery.
- Always cells have higher power density.
- Power density decreases during discharge.
- The unit of power density is Wh/kg

5. Energy efficiency:

- Energy efficiency for the secondary battery is given by Energy released during discharge

$$\% \text{ Energy efficiency} = \frac{\text{Energy released during discharging} \times 100}{\text{Energy consumed during charging}}$$

- Higher the energy efficiency better is the battery.

CHARACTERISTICS OF BATTERY

6. Cycle life:

- It is the number of charge and discharge cycles that are possible before failure occurs.
- Reasons for lowering of cycle life are
- Loss of electroactive material from the electrode due to rapid charging conditions.
- Corrosion of the contact wire.
- Irregular growth on the electrodes leading to short circuit.
- A good battery shall have high cycle life.

7. Energy density:

- This refers to the energy available from a cell to its weight.
- Energy density = Energy available from the cell / Weight of the cell

CHARACTERISTICS OF BATTERY

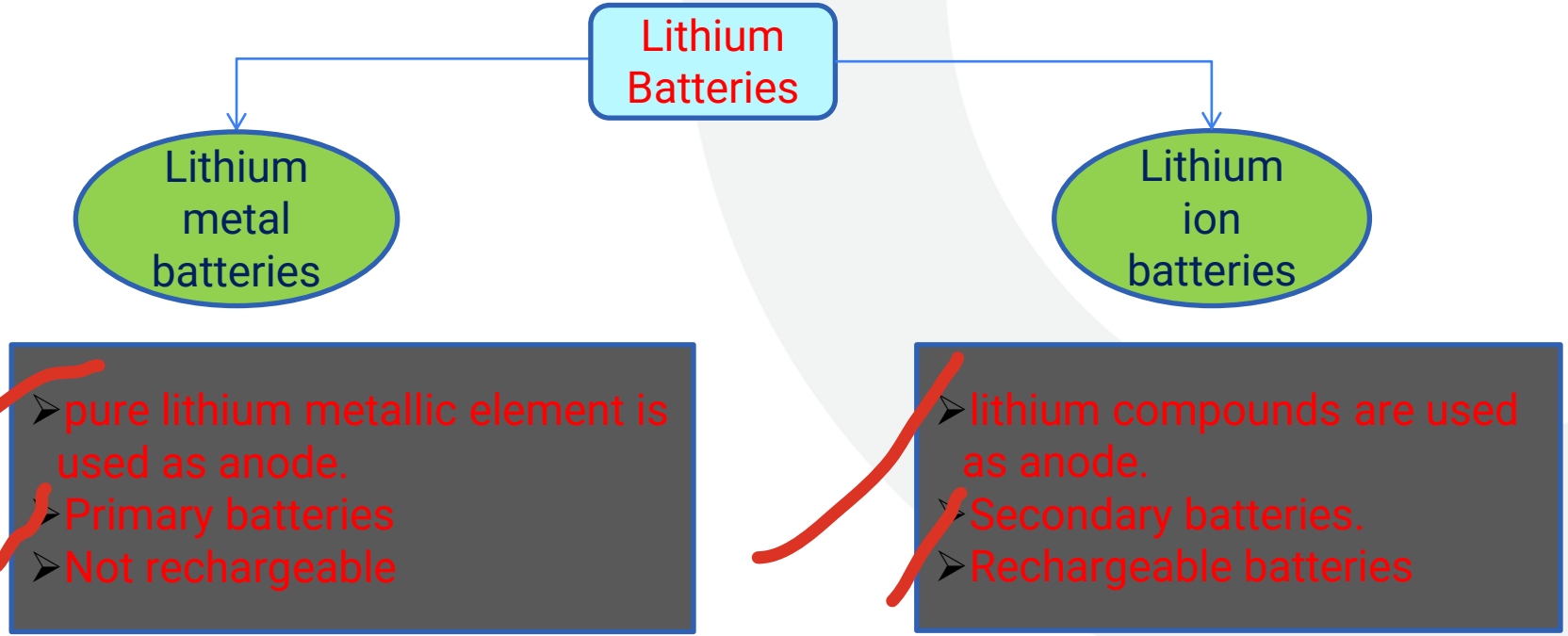
8. Shelf life:

- The length of time a battery can remain in storage without losing its capacity.
- Shelf life is the period of storage under specified conditions during which a battery retains its performance level.
- The reasons for shortening of the shelf life is due to self discharge.
- Self discharge is defined as the loss of useful capacity of the battery due to internal action within the cell.
- Lower the self discharge, longer the shelf life.

LITHIUM BATTERIES

- Lithium is the lightest of metals and it can float on water.
- The electrochemical properties of lithium are excellent, and it is also a highly reactive material.
- These properties gives Lithium the potential to achieve very high energy and power densities in high-density battery applications such as automotive and stand by power.
- Lithium batteries are primary batteries in which lithium metal (or) lithium compound acts as an Anode. A lithium cell can produce voltage from 1.5 V to about 3 V based on the types of materials used.

TYPES OF LITHIUM BATTERIES

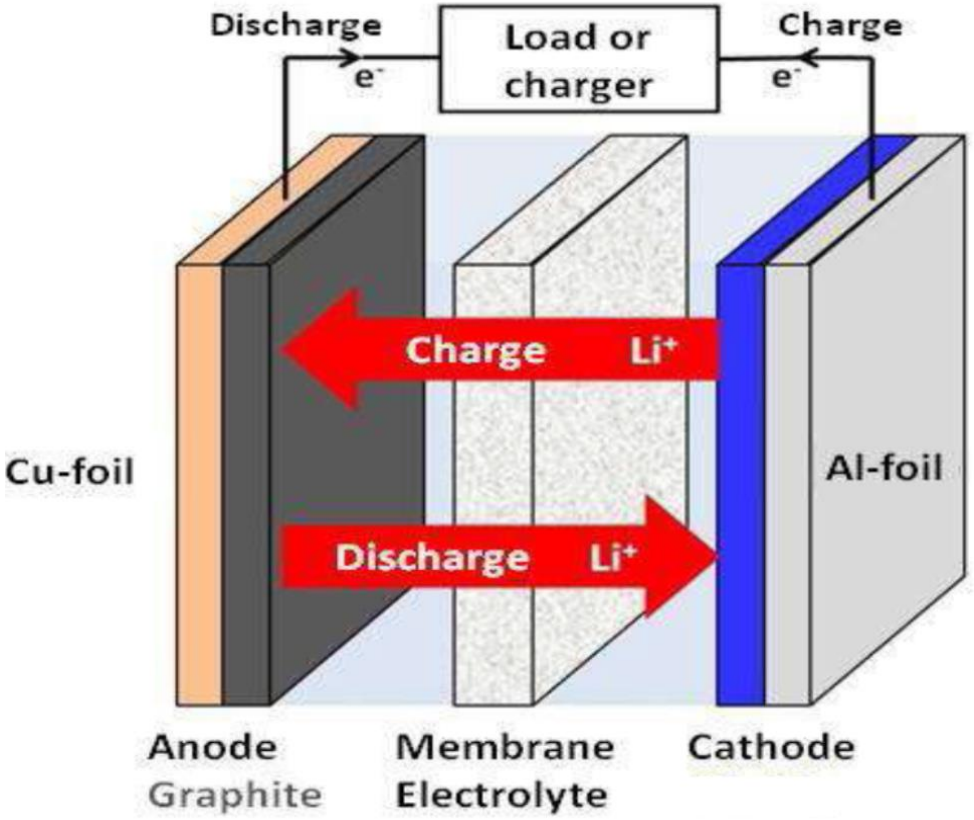


$LiCoO_2$ Battery

Construction:

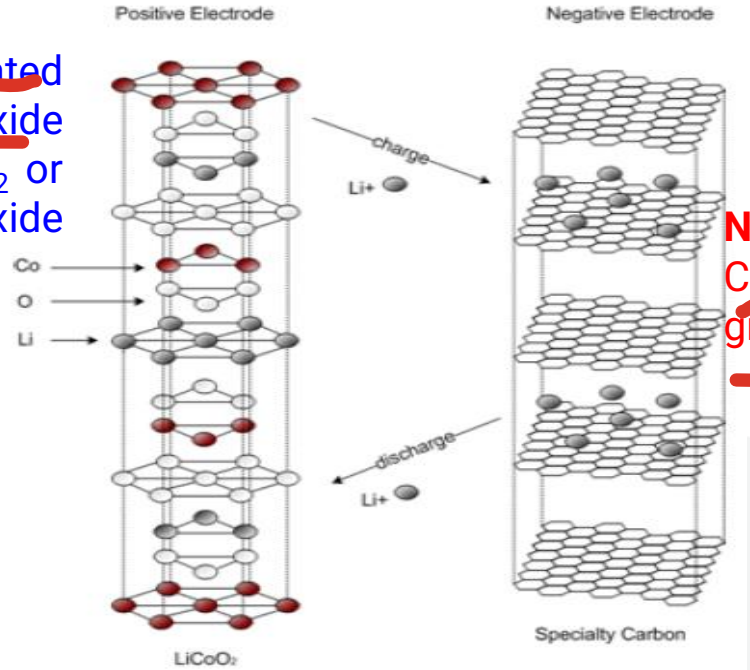
- Anode (-Ve electrode): Lithium metal dissolved into a carbon.
- Cathode(+Ve electrode): The cathode material is made up from Lithium liberating compounds, typically the three electro-active oxide materials,
  - Lithium Cobalt-oxide ($LiCoO_2$)
  - Lithium Manganese-oxide ($LiMn_2O_4$)
  - Lithium Nickel-oxide ($LiNiO_2$)
- Electrolyte: Lithium salts dissolved in organic solvents. Electrolyte should help in effective conduction of Li-ions to cathode during discharge (ionic conductivity rather than electronic conductivity).
- Seperator: A fine porous polymer film.

$LiCoO_2$ Battery



$LiCoO_2$ Battery

Positive electrode: Lithiated form of a transition metal oxide (lithium cobalt oxide-$LiCoO_2$ or lithium manganese oxide $LiMn_2O_4$)



Negative electrode: Carbon (C), usually graphite (C_6) with lithium

Electrolyte: solid lithium-salt electrolytes ($LiPF_6$, $LiBF_4$, or $LiClO_4$) and organic solvents (ether)

$LiCoO_2$ Battery

Working :

The chemical reaction that takes place inside the battery is as follows, during charge and discharge operation:

During discharge:

At Anode: Li ions present in graphite layer will be oxidized. Electrons will flow through external circuit to cathode and lithium ions flow through the organic electrolyte towards cathode.

$$Li_x\text{-}C_6 \longrightarrow xLi^+ + 6C + xe^-$$

At Cathode: Li-ions are reduced to Li and are inserted into the layered structure of metal oxide.

$$Li_{1-x}CoO_2 + xLi^+ + xe^- \longrightarrow LiCoO_2$$

$LiCoO_2$ Battery

During charging:

At Cathode: Li ions present in layered structure of $LiCoO_2$ will be oxidized. Electrons will flow through external circuit to ~~cathode~~ ANODE and lithium ions flow through the organic electrolyte towards ~~cathode.~~

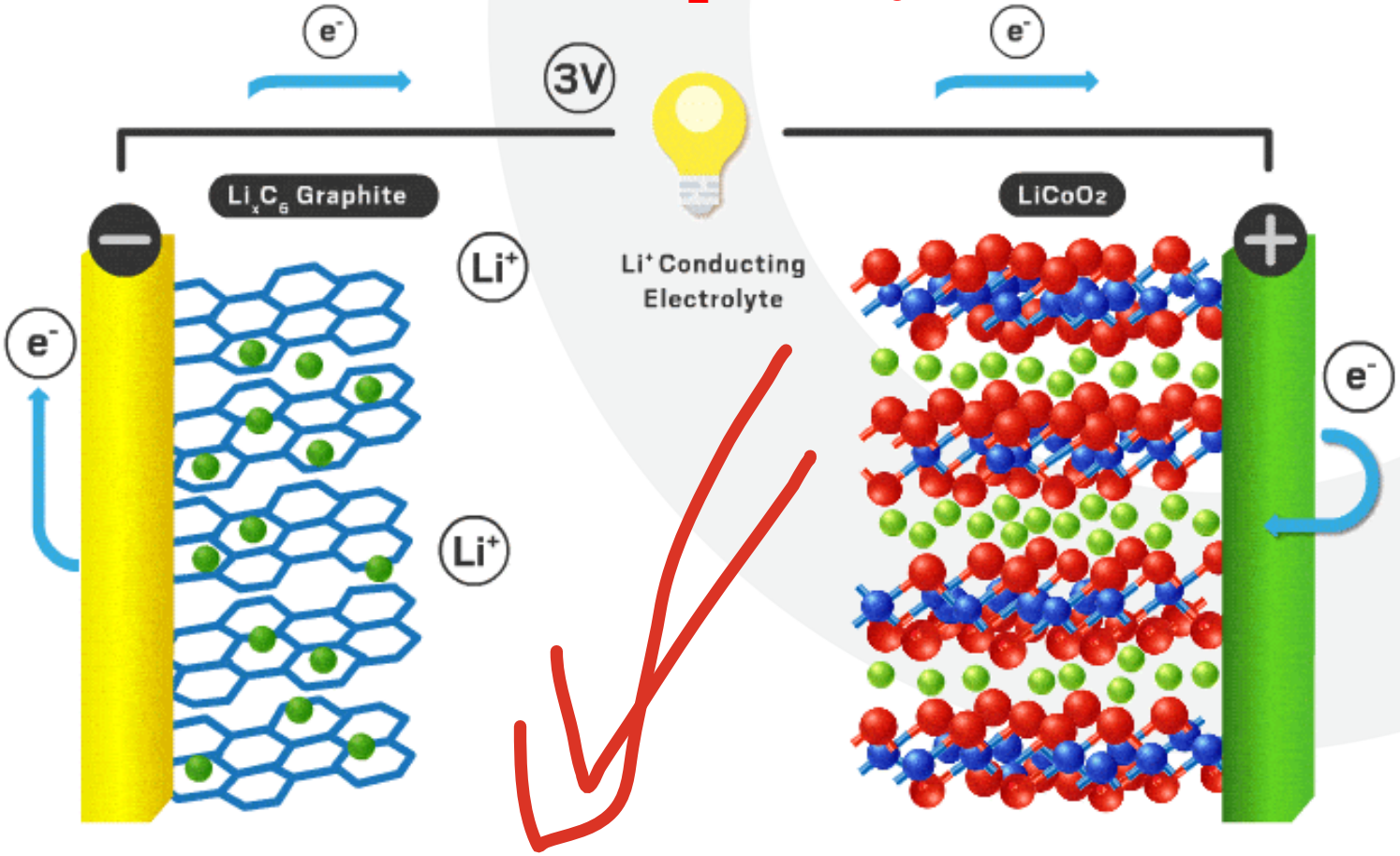
$$LiCoO_2 \longrightarrow Li_{1-x}CoO_2 + xLi^+ + xe^-$$

At anode: Li-ions are reduced to Li and are inserted into the layered structure of graphite.

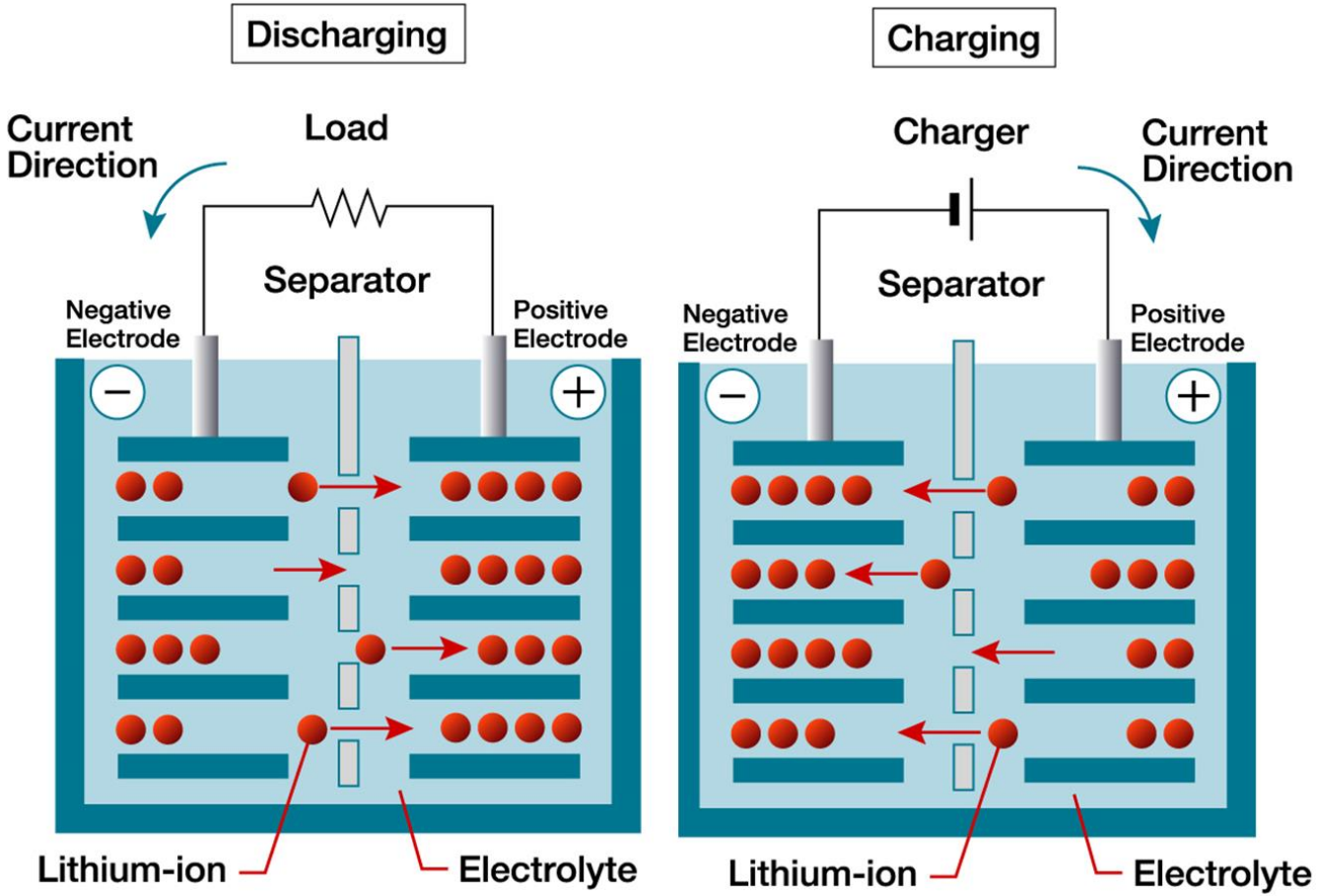
$$xLi^+ + 6C + xe^- \longrightarrow Li_x\text{-}C_6$$

$$LiCoO_2 + C_n \underset{\text{discharge}}{\overset{\text{charge}}{\rightleftharpoons}} C_nLi_x + Li_{1-x}CoO_2$$

$LiCoO_2$ Battery

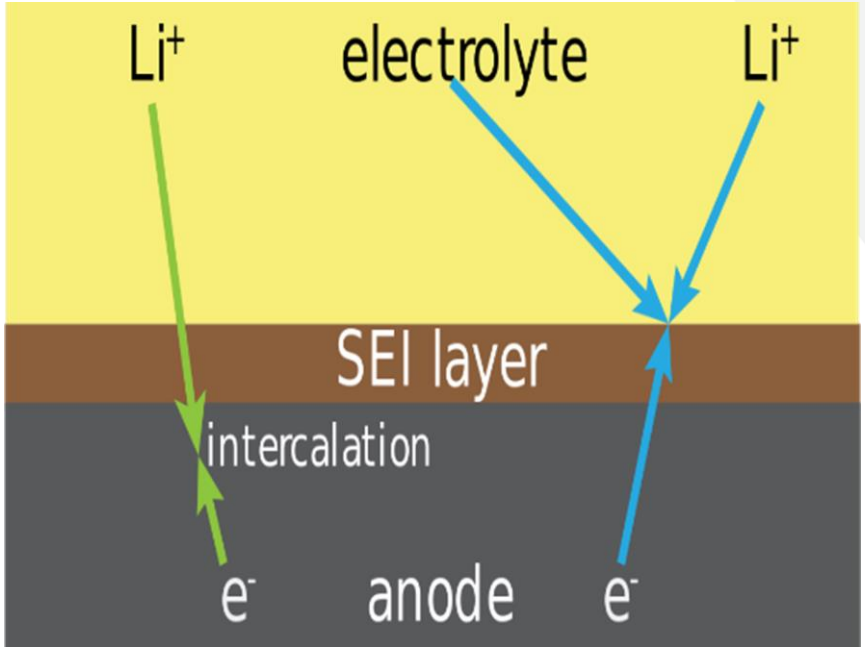


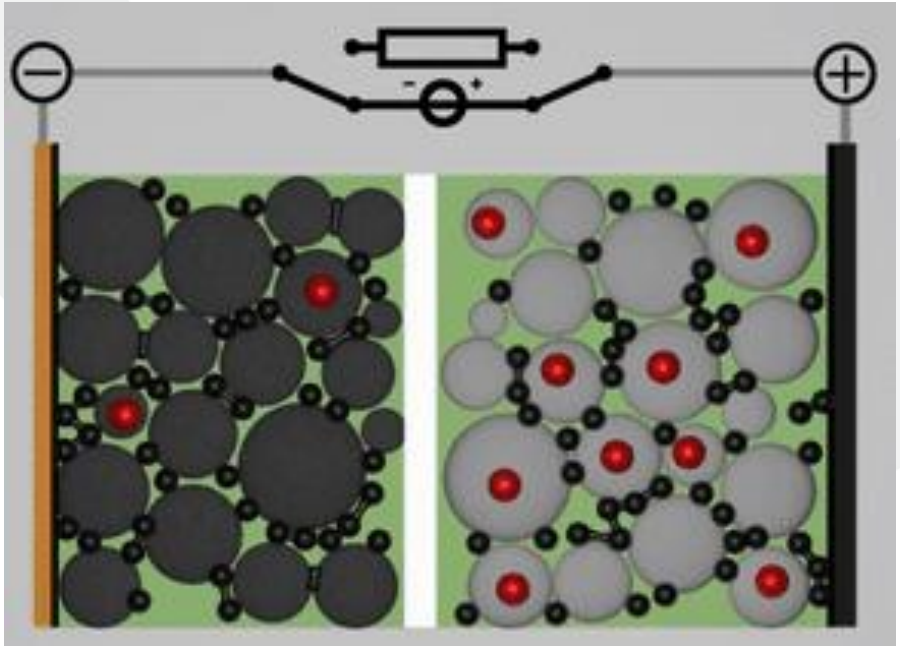
$LiCoO_2$ Battery



Importance of SEI layer

The solid electrolyte interface (SEI) is a passivation layer formed on the surface of lithium-ion battery (LIB) anode materials produced by electrolyte decomposition. The quality of the SEI plays a critical role in the cyclability, rate capacity, irreversible capacity loss and safety of lithium-ion batteries (LIBs).





$LiCoO_2$ Battery

Advantages

- They have high energy density than other rechargeable batteries
- They are less weight
- They produce high voltage out about 4 V as compared with other batteries.
- They have improved safety, i.e., more resistance to overcharge
- Fast charge and discharge rate.

Disadvantage

- They are expensive

Lithium air Batteries

- The lithium–air battery (Li–air) is a metal–air electrochemical cell or battery chemistry that uses oxidation of lithium at the anode and reduction of oxygen at the cathode to induce a current flow.

- The anode material is Lithium metal.

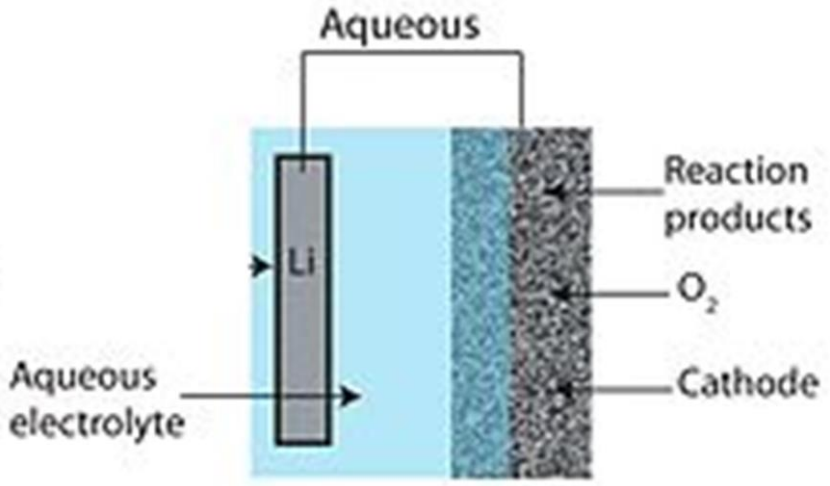
- The cathode is porous carbon with metal catalysts, filled with oxygen gas.

- The aqueous electrolytes: Lithium salts dissolved in water. It may be acidic/alkaline electrolyte.

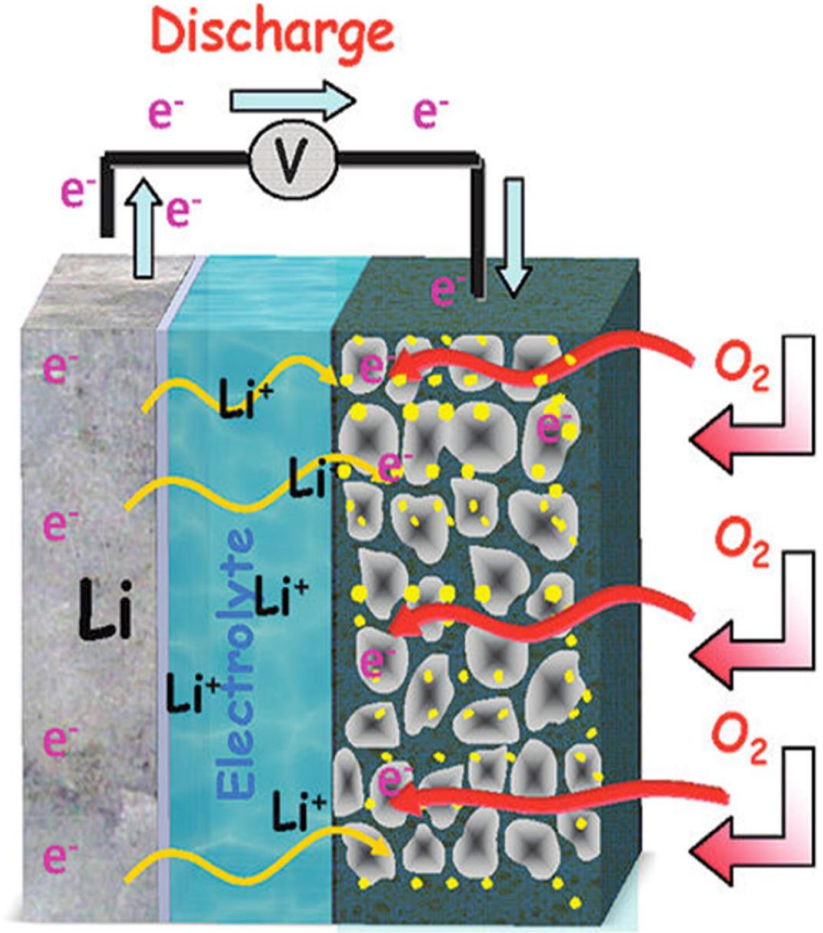
$$Li^{(s)} \leftrightarrow Li^+ + e^- \text{ (anode)}$$

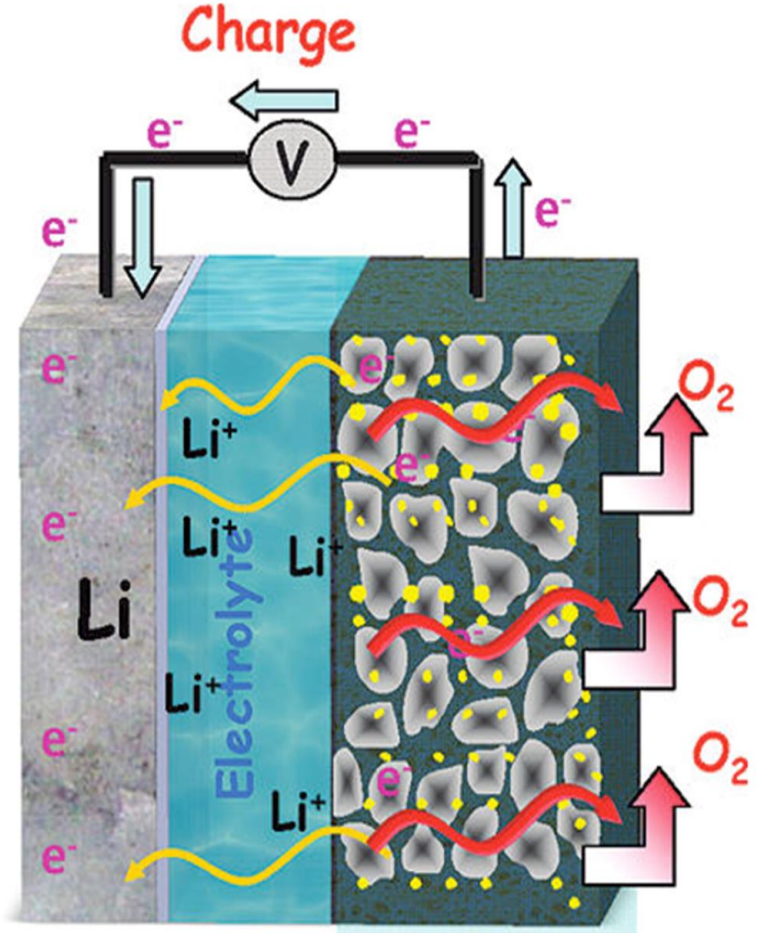
$$Li^+ + 1/2O_2 + e^- \leftrightarrow 1/2Li_2O_2 \text{ (cathode)}$$

$$Li^+ + 1/4O_2 + e^- \leftrightarrow 1/2Li_2O \text{ (cathode)}$$



Lithium air Batteries

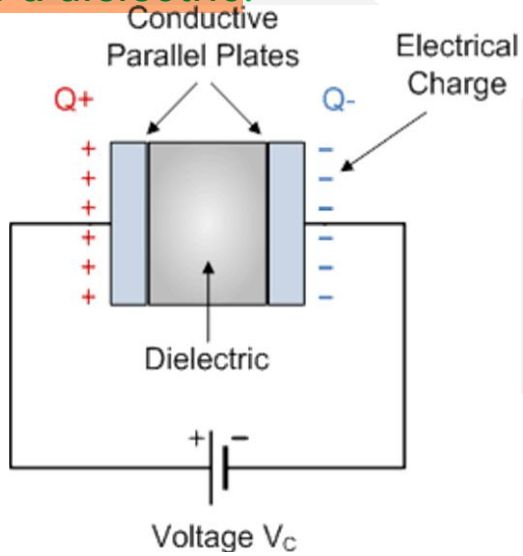


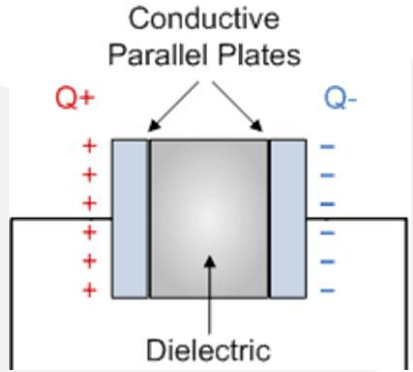


Capacitors

- A capacitor is a two-terminal electrical device that can store energy in the form of an electric charge.
- It consists of two electrical conductors that are separated by a distance.
- The space between the conductors may be filled by vacuum or with an insulating material known as a dielectric.





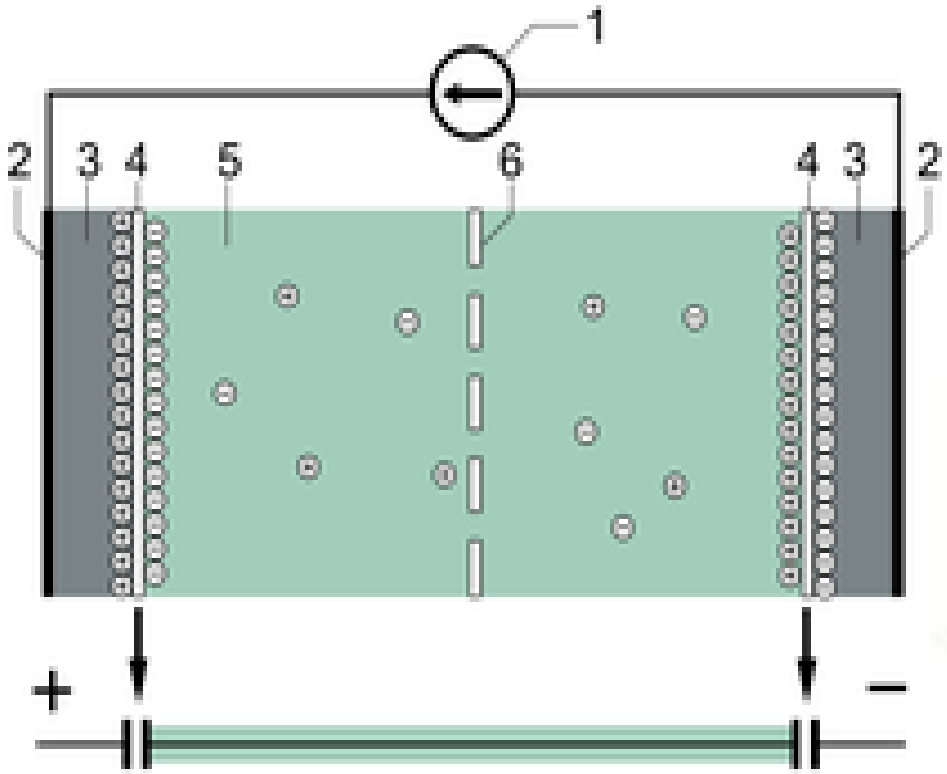


Super Capacitors

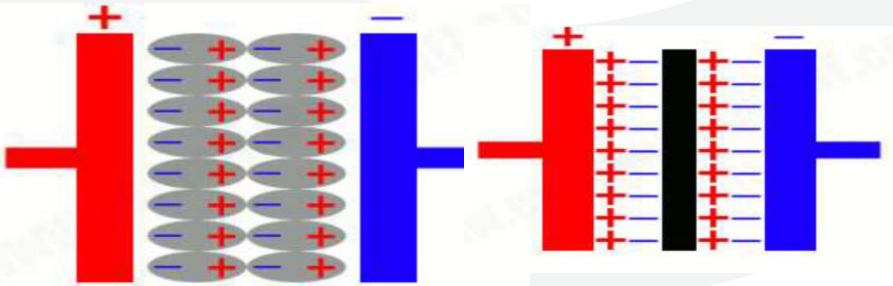
- A supercapacitor also called an ultracapacitor, is a high-capacity capacitor, with a capacitance value much higher than other capacitors but with lower voltage limits.
- It bridges the gap between electrolytic capacitors and rechargeable batteries.
- It typically stores 10 to 100 times more energy per unit volume or mass than electrolytic capacitors, can accept and deliver charge much faster than batteries, and tolerates many more charge and recharge cycles than rechargeable batteries.
- Unlike ordinary capacitors, supercapacitors do not use the conventional solid dielectric, but rather, they use electrostatic double layer capacitance and electrochemical pseudo capacitance, both of which contribute to the total capacitance of the capacitor, with a few difference

Super Capacitors

Construction of a supercapacitor:

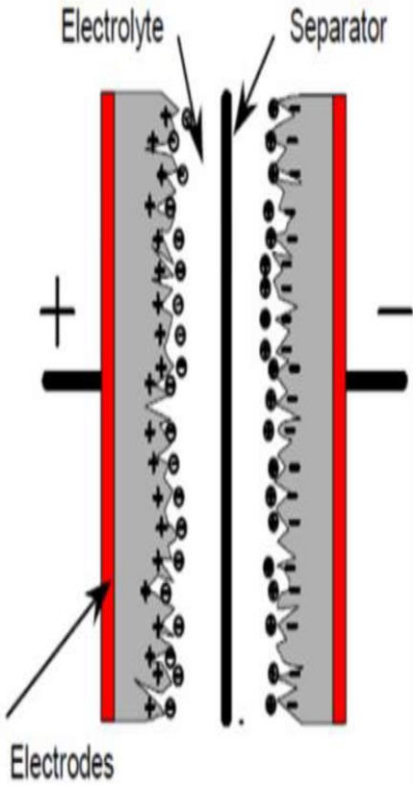


(1) power source,
(2) collector,
(3) polarized electrode,
(4) Helmholtz double layer,
(5) Electrolyte having positive and negative ions,
(6) Separator.



Super Capacitors

- Supercapacitors are constructed with two metal foils, each coated with an electrode material such as activated carbon.
- The electrodes are kept apart by an ion permeable membrane (separator) used as an insulator to protect the electrodes against short circuits.
- The construction is subsequently rolled or folded into a cylindrical or rectangular shape and is packed in an aluminium can.
- When the plates are charged, an opposite charge forms on either side of the separator, creating an electric **double layer.** (Double-layer capacitors)



Super Capacitors

Commonly used materials

- Carbon nano tubes, carbon aerogels are used for supercapacitors plates or electrodes.
- Sodium perchlorate ($NaClO_4$) or lithium perchlorate ($LiClO_4$) are used as electrolytes.
- Polyacrylonitrile$(C_3H_3N)_n$ is used as a separator(thickness 0.3-0.8 nm).
- Aluminium as a packing component.

Why these materials are taken

- Electrodes:

1)Carbon nanotubes greatly improve capacitor performance, due to the highly wettable surface area and high conductivity. 2)Highly porous.

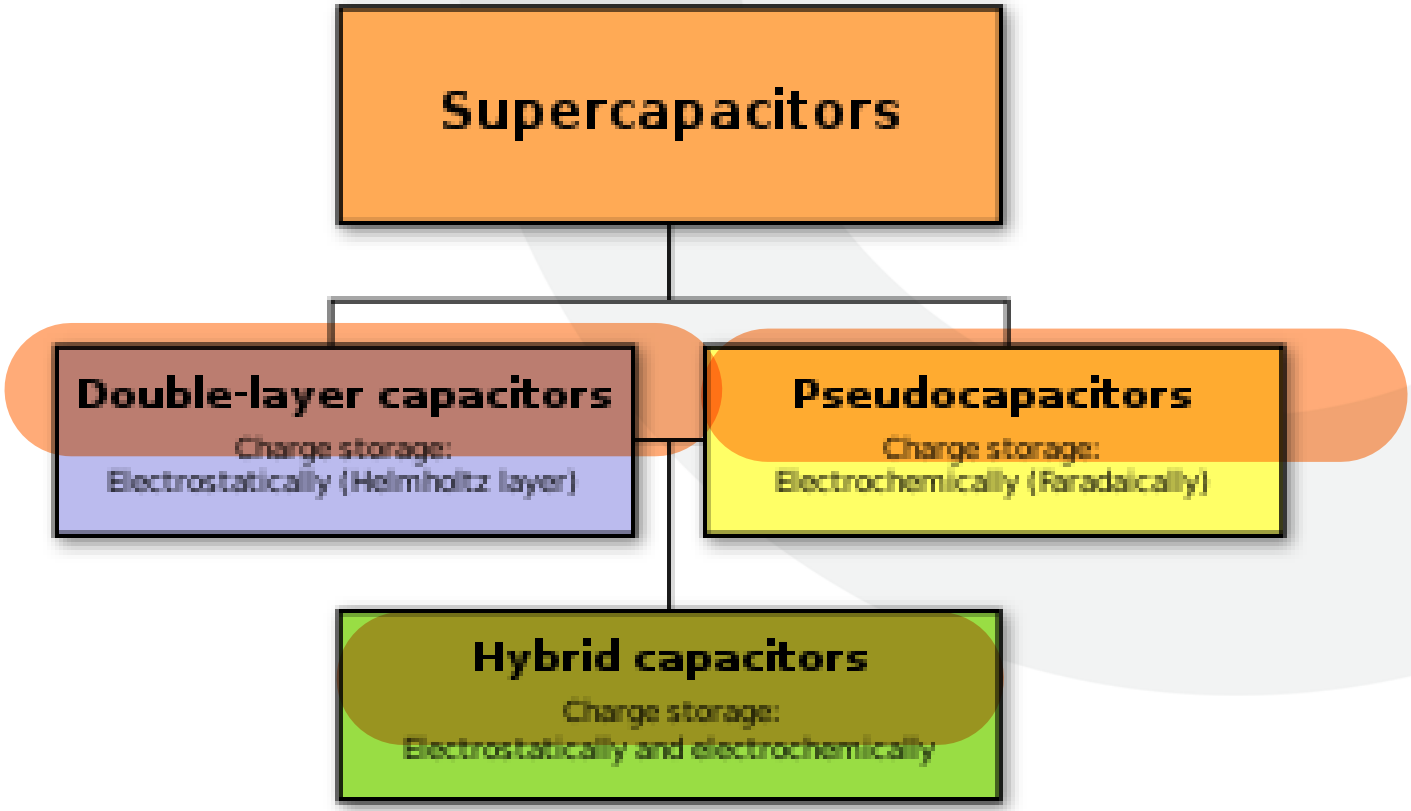
- Electrolytes:

1) Wide working temperature (-90^0c to 400^0c). 2) Nonflammable and low toxic. 3)Non-corrosive to electrode & packing components.

- Separator:

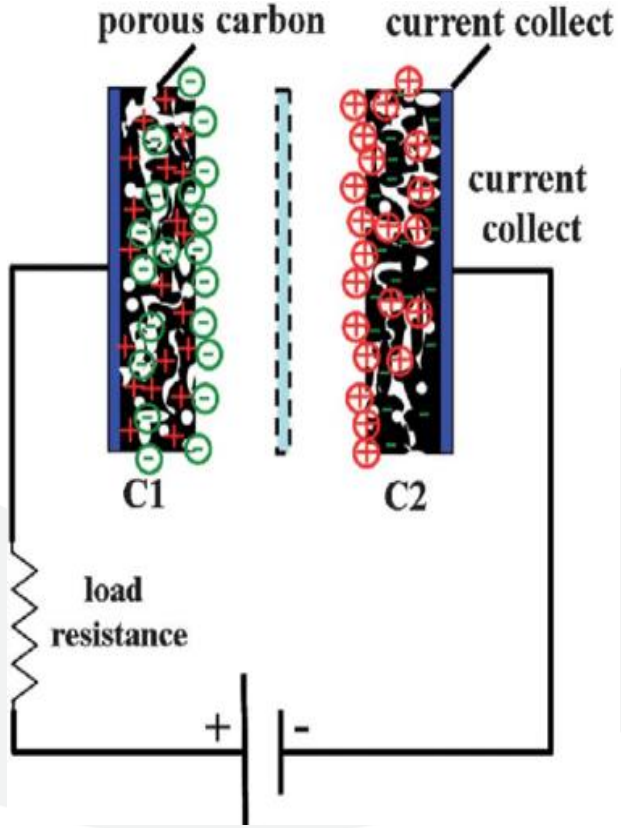
1) Unique tensile strength (103MegaPascals). 2)Electrical conductivity ($1.5x10^4$ S/m). 3)Not degraded easily.

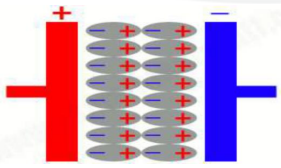
Classification of Super Capacitors



Electrical double layer Super Capacitors

- EDLCs store charge electrostatically and there is no transfer of charge between electrode and electrolyte.
- EDLCs utilize an electrochemical double-layer of charge to store energy. As voltage is applied, charge accumulates on the electrode surfaces.
- The subclasses of EDLCs are distinguished primarily by the form of carbon they use as an electrode material.
- Different forms of carbon materials that can be used to store charge in EDLC electrodes are activated carbons, carbon aerogels, and carbon nanotubes.

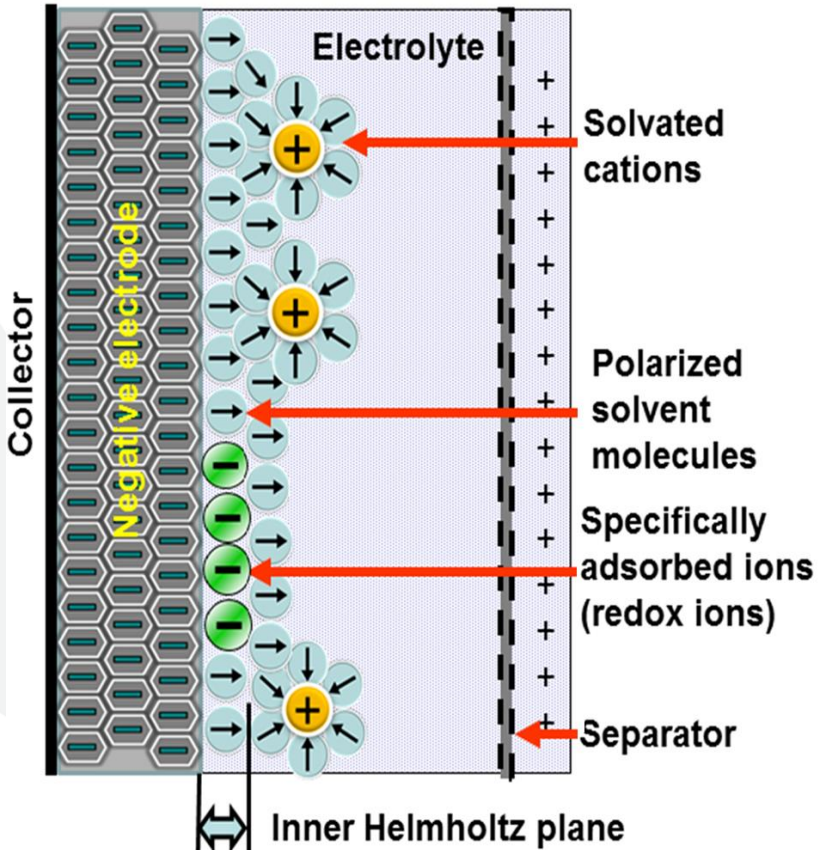




Pseudo Capacitors

- In contrast to EDLCs, which store charge electrostatically, pseudocapacitors store charge Faradically through the transfer of charge between electrode and electrolyte. This is accomplished through reduction-oxidation reactions.

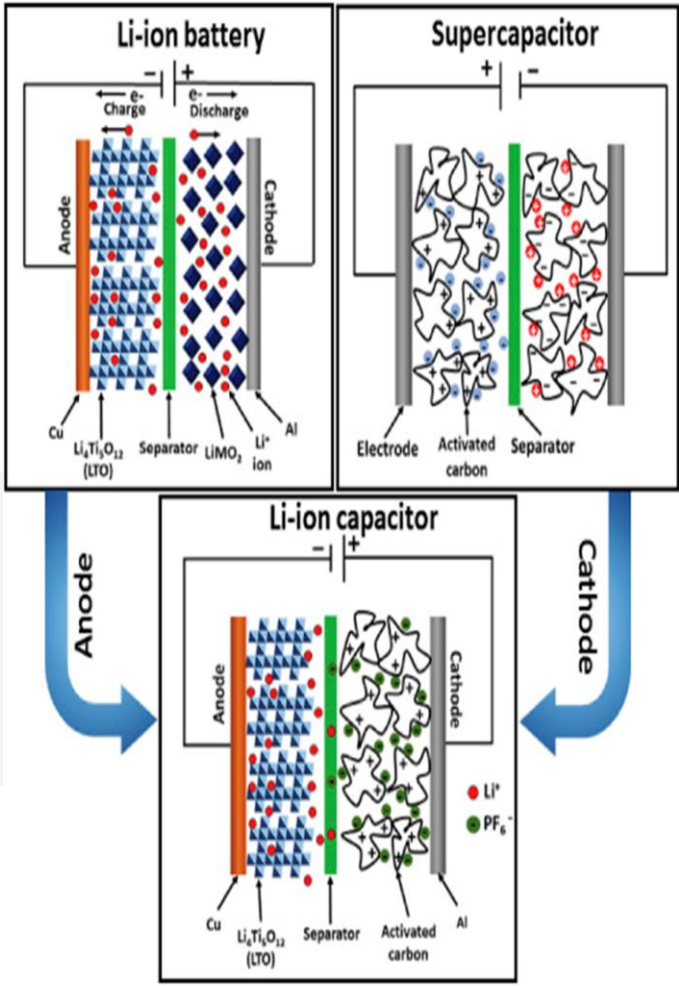
- Faradic processes may allow pseudocapacitors to achieve greater capacitances and energy densities than EDLCs. There are two electrode materials that are used to store charge in pseudo capacitors, conducting polymers and metal oxides.



Hybrid Capacitors

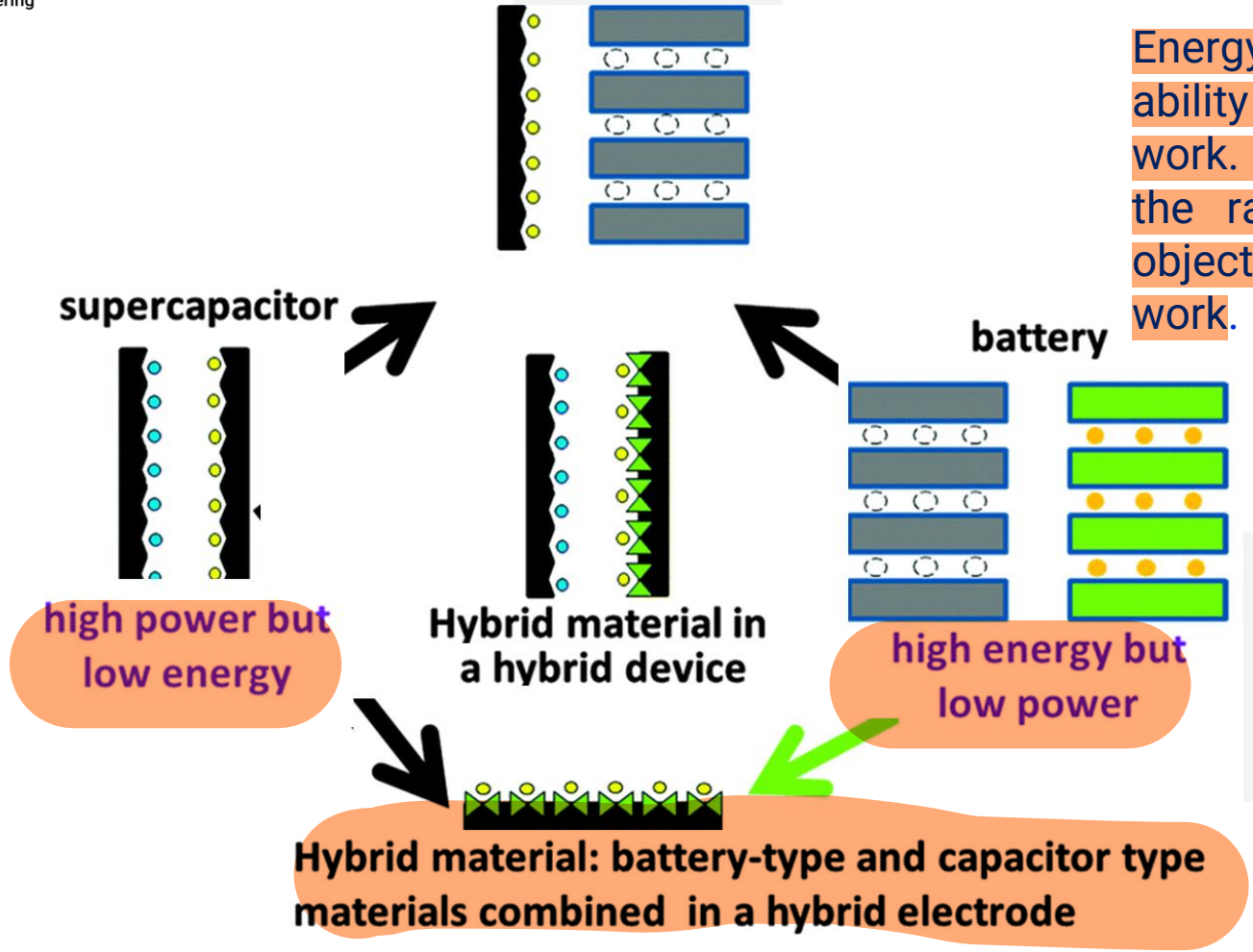
- Hybrid capacitors attempt to exploit the relative advantages and mitigate the relative disadvantages of EDLCs and pseudo capacitors to realize better performance characteristics.

- Utilizing both Faradic and non-Faradic processes to store charge, hybrid capacitors have achieved energy and power densities greater than EDLCs without the sacrifices in cycling stability and affordability that have limited the success of pseudo capacitors.

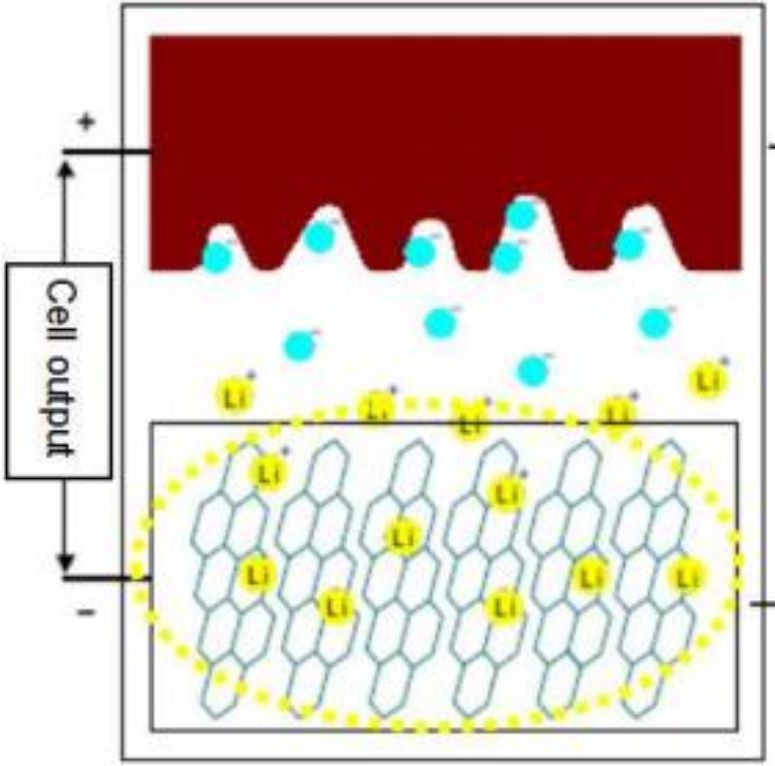


Hybrid Capacitors

Energy is an object's ability to complete work. Power refers to the rate at which an object can complete work.



Hybrid Capacitors



Positive Electrode: electric double layer with carbon electrode

Negative Electrode: dope/ un-dope reaction of Li-ion in carbon like material

ADVANTAGES of super capacitor

- High energy storage.
- Wide working temperature(-400c to 600c).
- Eco-friendly.
- Quick charging time.
- Maximum life cycle.
- High cycle efficiency (95%).
- Extremely low internal resistance.
- Safe.

Limitations of super capacitor

- Low energy density; usually holds 1/5 – 1/10 of a battery.
- Cannot use the full energy spectrum for some applications.
- The voltage varies with the energy stored.
- Have high self-discharge rate.
- Individual cells have low voltages, and so serial connections are needed to obtain higher voltages.
- Requires expert electronic control.
- Cannot be used in AC and high frequency circuits.
- High cost.

Rechargeable Batteries Vs. Supercapacitors

Supercapacitors

- Higher power density
- Much faster charge and discharge rate
- Environmentally friendly
- Extremely low internal resistance.
- High efficiency (97-98%)
- Over a million charge-discharge cycles

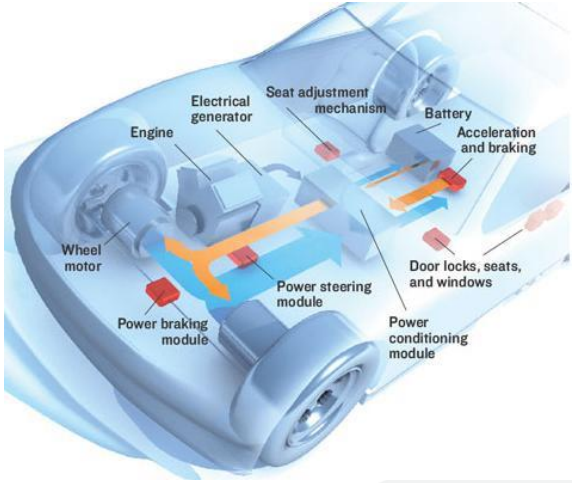
Batteries

- Have higher energy density
- Typically, 200–1000 charge-discharge cycles
- Contain highly reactive and hazardous chemicals
- Negatively effected by low temperatures

Applications for Supercapacitors



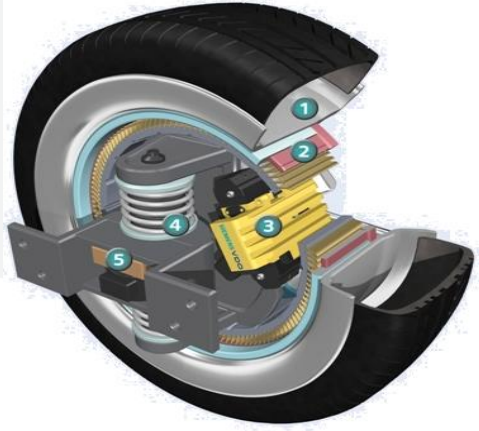
- Back up for uninterruptable power supplies (UPS)
- Light weight power supplies for small aircraft



- Provide short duration power for various vehicle systems such as breaking or steering



- Used to absorb power during short periods of generation such as Regenerative Braking
- Extend range and battery life in Hybrid Electric Vehicles (HEV)



RV College of Engineering®

Bus operated on Supercapacitor in Shanghai







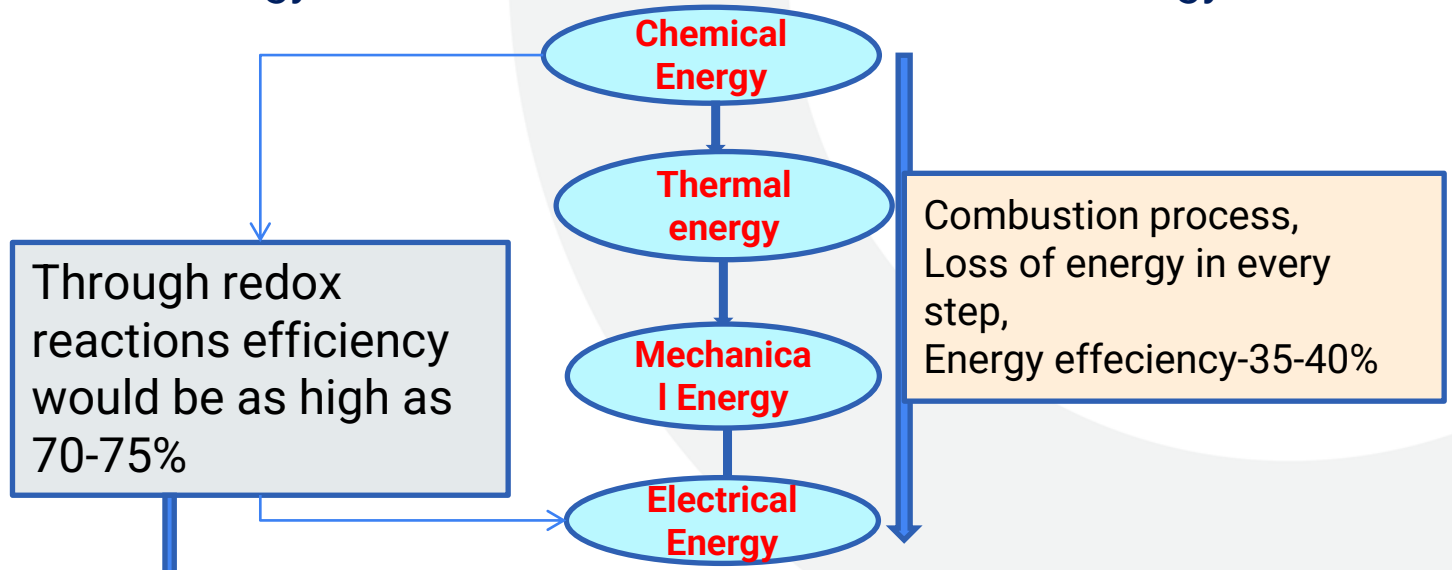
Supercapacitor installed in bus

Supercapacitors

FUEL CELLS

Fossil fuels are conventional energy sources.

- The chemical energy of a fuel is converted into electrical energy as follows.



FUEL CELLS

Representation of Fuel Cells

Fuel/ Electrode/ Electrolyte/ Electrode/ Oxidant



Cell Reactions:

At Anode:

Fuel → Oxidised Product + ne

At Cathode:

Oxidant + ne → Reduced Product

Overall cell reaction:

Fuel + Oxidant → Oxidised product + Reduced Product

Special Features of FUEL CELLS

- Fuel cells consists of two catalytically active electrodes, oxidant and a reductant.
- The fuel and oxidants are not stored in the cell.
- The thickness of the cell may be less than one cm.
- In a fuel cell, redox reaction occurs involving fuel and oxidant.
- Fuel cells are energy conversion devices.
- Eco friendly devices.
- Fuel undergoes oxidation whereas oxidants undergo reduction.

FUEL CELLS

Sl No	Battery	Fuel cell
1	These are the energy storage devices	These are energy conversion devices
2	Secondary batteries are rechargeable	Fuel cells are not chargeable.
3	The reactants and products form integral part of the batteries	In fuel cells, there is a continuous supply of fuel, oxidant
4	Products remain in the cell	Products are continuously removed from the cell

FUEL CELLS

Advantages of fuel cells

- Savings in fossil fuels due to the high efficiency of electrochemical energy conversion.
- Low pollution level, no noxious exhaust gases formed.
- Low noise level, Low maintenance, exchangeable parts.
- No need of charging.
- Theoretically, the efficiency can be 100%. In practice, the efficiency is 50-80% which is high compared to conventional methods.
- Have few moving parts and thus require minimal maintenance, reducing life cycle costs of energy production.

Hydrogen-Oxygen FUEL CELLS

- The hydrogen oxygen fuel cell is categorized under alkaline fuel cells (AFC).
- The alkaline fuel cell uses an alkaline electrolyte such as 40% aqueous potassium hydroxide.
- In alkaline fuel cells, negative ions travel through the electrolyte to the anode where they combine with hydrogen to generate water and electrons.

Hydrogen-Oxygen FUEL CELLS

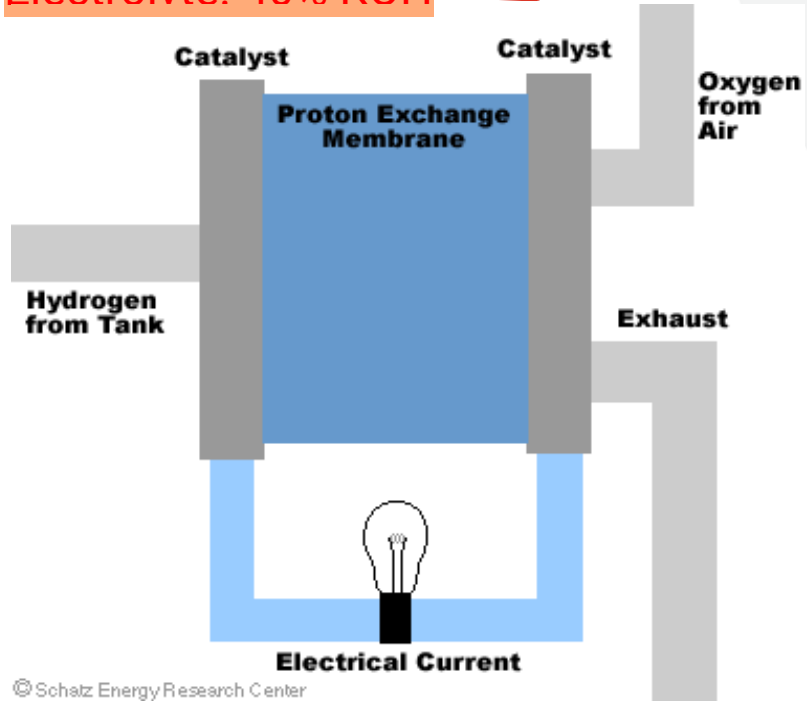
Anode: porous carbon with Pt

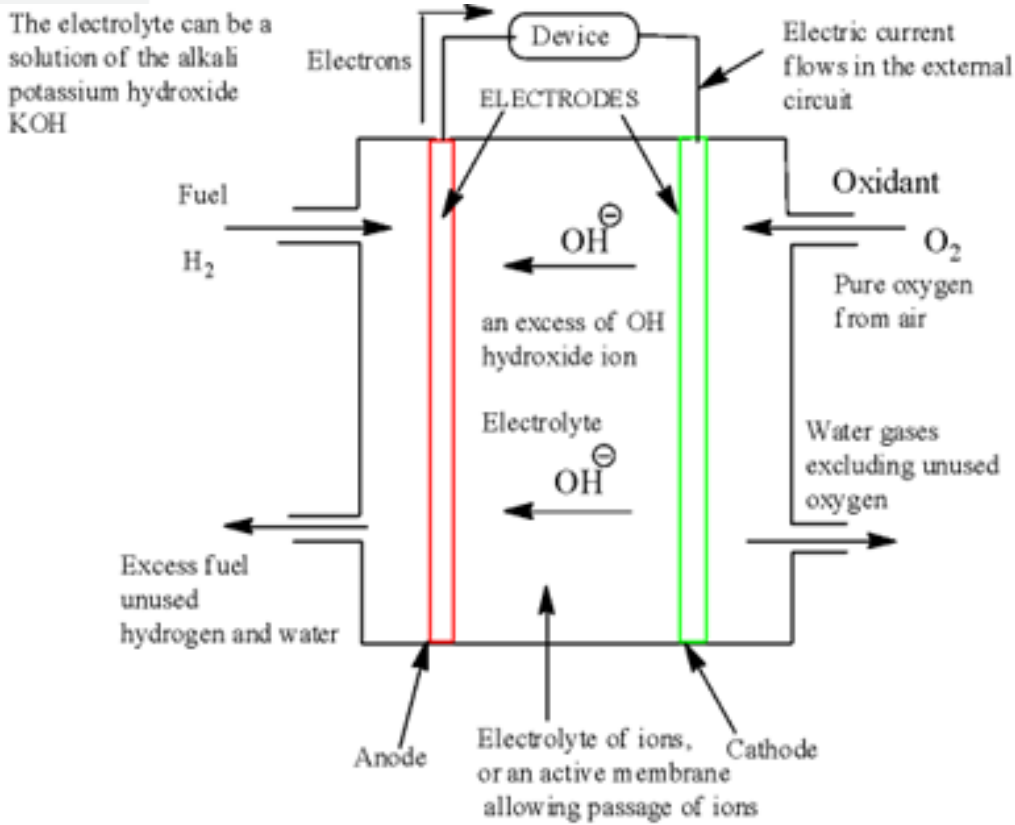
Fuel: Hydrogen

Cathode: porous carbon with Ag

Oxidant: Oxygen

Electrolyte: 40% KOH





Hydrogen-Oxygen FUEL CELLS

Working:

At anode:

Hydrogen gas pumped at the anode surface the following oxidation reaction occurs:

$H_2(g) + 4OH^-(aq) \longrightarrow 4H_2O(l) + 4e^-$

At cathode:

The liberated electrons from anode then passes through the external circuit where work is performed, & then finally to the cathode

$O_2(g) + 2H_2O(l) + 4e^- \longrightarrow 4OH^-(aq)$

Net reaction:

$H_2(g) + 2O_2(g) \longrightarrow 2H_2O(l)$

Hydrogen-Oxygen FUEL CELLS

Advantages:

- **Zero emissions:** The only byproduct of a fuel cell is water vapor.
- **High efficiency:** Without using combustion, fuel cells transform chemical energy directly into electricity. Fuel cells can thus reach significant energy conversion efficiency.
- **High power density:** Fuel cells can be used in applications where space is limited because of their high-power density.
- **Replaceable:** Because fuel cell systems are modular, it is simple to swap out individual pieces.
- Water is produced by a hydrogen oxygen fuel cell and is of portable quality.
- Fuel cell systems: no recharging.

Hydrogen-Oxygen FUEL CELLS

Limitations:

- It is difficult to manufacture and stores a high pure hydrogen
- It is very expensive as compared to battery
- High cost of pure hydrogen
- Life span of the cells are not accurately known
- Lack of infrastructure for distributing hydrogen.
- Liquification of hydrogen requires 30% of the stored energy

Methanol-Oxygen FUEL CELLS

- Methanol is used as the fuel in direct-methanol fuel cells, which are a type of proton-exchange fuel cell.

- Their key benefit is the simplicity of transporting methanol, an environmentally stable and energy-dense liquid.

Construction:

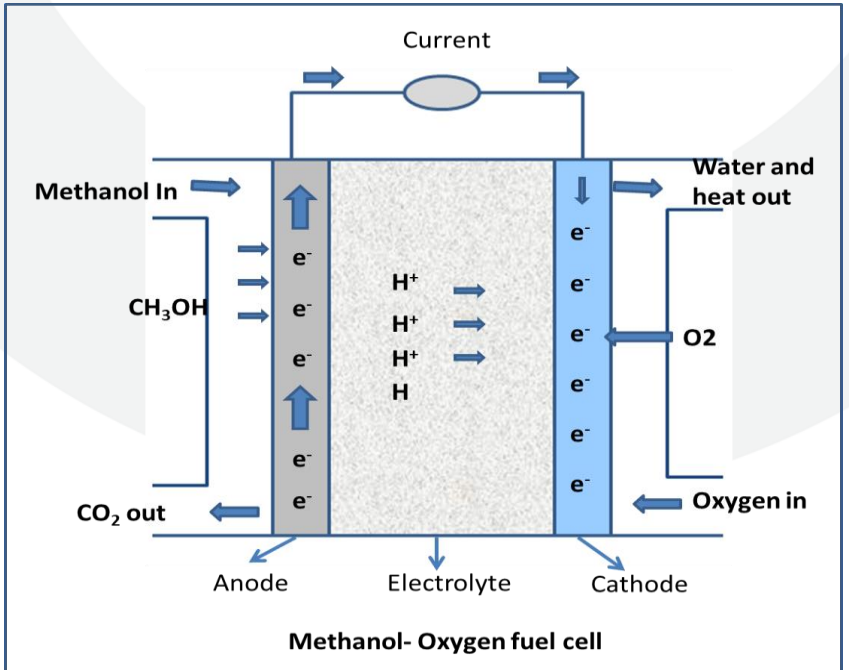
- **Anode**: Porous Ni / Pt-Pd Catalyst. **Fuel at anode:** Methanol

- **Cathode:** Porous Ni / Ag catalyst. **Oxidant at cathode:** Oxygen

- **Electrolyte used:** H_2SO_4

- **Membrane used:** Proton exchange membrane (PEM)

- **Operating temperature:** 50-120^{0}C



Methanol- Oxygen fuel cell

Methanol-Oxygen FUEL CELLS

Working mechanism

- Fuel cell is fed with methanol and water at anode site. After the injection of methanol, it passes through anode catalyst (Pt/Rh) and membrane, then splits into carbon dioxide and proton with liberation of electron.
- Carbon dioxide escapes from the system and the electron will move in the external circuit.
- The electron reacts with oxygen at cathode in presence of cathode catalyst (Ag) to produce water as useful biproduct.

At Anode

$$CH_3OH + H_2O \rightarrow CO_2 + 6H^+ + 6e^-$$

At Cathode

$$3/2\ O_2 + 6H^+ + 6e^- \rightarrow 3H_2O$$

Overall reaction

$$CH_3OH + 3/2\ O_2 \rightarrow CO_2 + 2H_2O$$

Methanol-Oxygen FUEL CELLS

- Alkali electrolyte are not preferred in advanced methanol fuel cell because, liberated CO_2 during reaction is converted into carbonate in presence of an alkali and would have an adverse effect on working of the cell.
- Hence, design of methanol-oxygen fuel cell in acid medium is attempted electrodes made of Pt-Ru-Ni and Pt-Ru-Ni alloys are used.
- CO_2 formed blocks the porosities as carbonates and efficiency of the cell is decreased.

$$CO_2 + 2OH^- \rightarrow CO_3^{-2} + H_2O$$

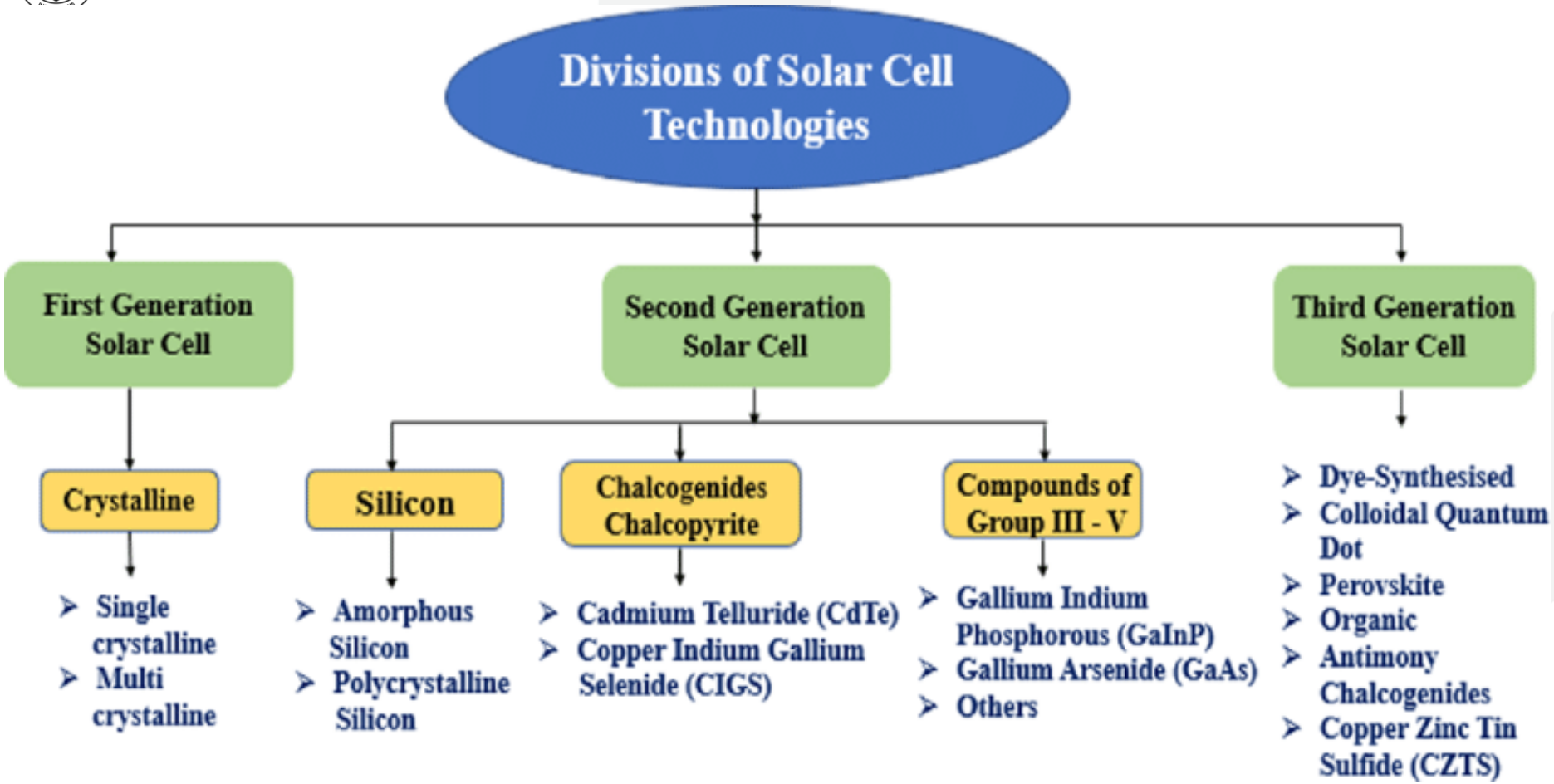
Methanol-Oxygen FUEL CELLS

- All kinds of portable, automotive and mobile applications like, Powering laptop, computers, cellular phones, digital cameras.
- Fuel cell vehicles (FCVs)
- Spacecraft applications
- Any consumables which require long lasting power compared to Li-ion batteries.

PHOTOVOLTAICS

- Solar energy is a renewable energy, and it is inexhaustible.
- The earth is receiving approximately 1.8×10^{11} MW of energy which is thousands of times larger than the requirement of energy for present consumption.
- Solar energy is the only energy that fulfils all our present and future energy needs if we utilize properly.
- Solar energy can be directly converted into heat energy or electricity.
- **A solar cell, or photovoltaic cell, is an electrical device that converts the light energy directly into electricity by the photovoltaic effect.**

Classification of Solar Cells



Advantages of Solar Cells

- Solar energy is renewable energy which can be continuously drawn from the sun.
- It is cheaper technology because once it is installed there will be minimum maintenance charges for small usage.
- Solar energy is environment friendly and green energy because it doesn't produce any greenhouse gasses and no pollutants.
- This technology does not involve any combustion reaction or radioactive residue hence no pollution.

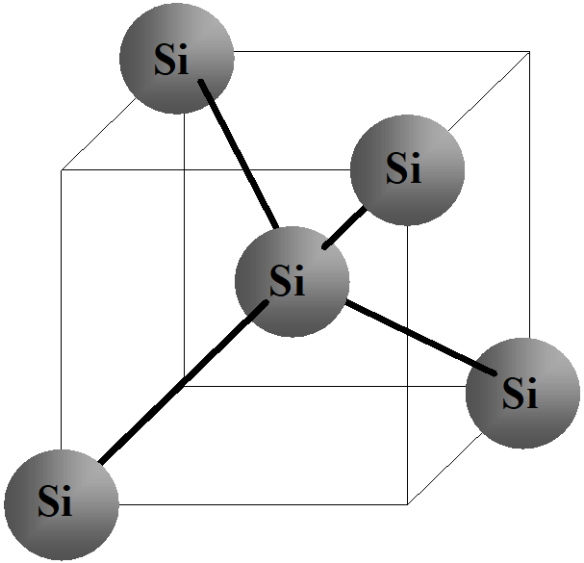
Disadvantages of Solar Cells

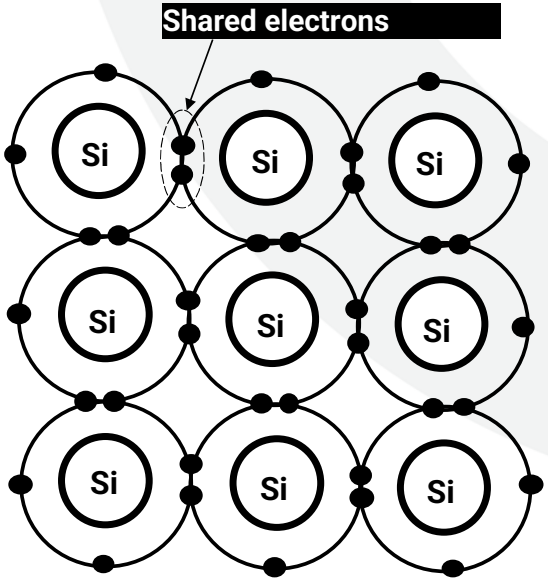
- Space utilization: Solar cells require large area for installation.
- It requires high investment cost for a big power plant.
- During the rainy season power production is less hence grid maintenance will be difficult.
- Solar energy is produced only in the daytime hence storage will be a big challenge.

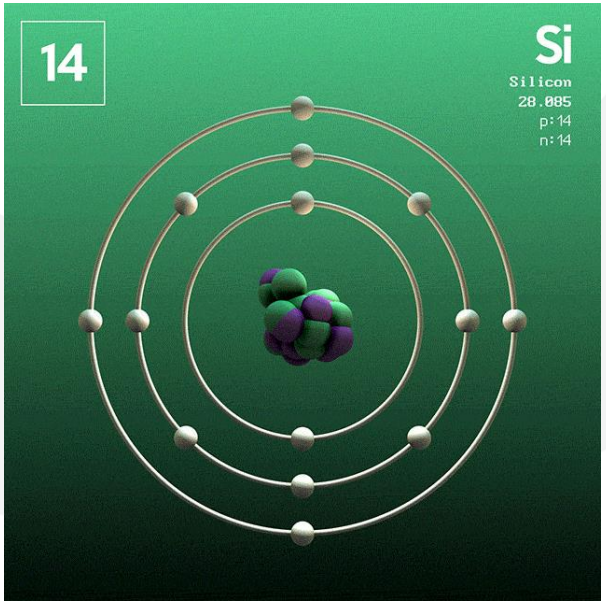
SILICON

- Silicon is group IV element – with 4 electrons in their valence shell.

- When silicon atoms are brought together, each atom forms covalent bond with 4 silicon atoms in a tetrahedron geometry.

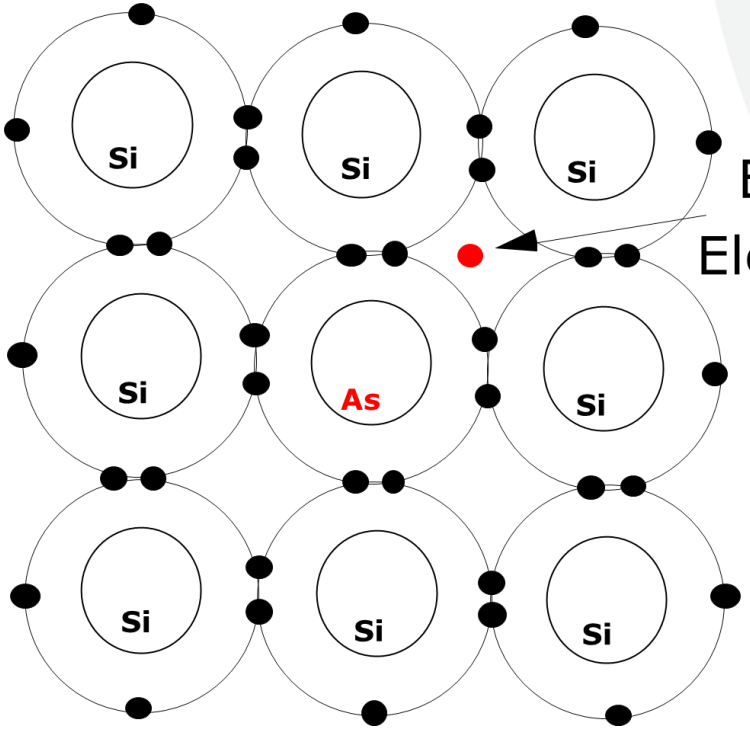


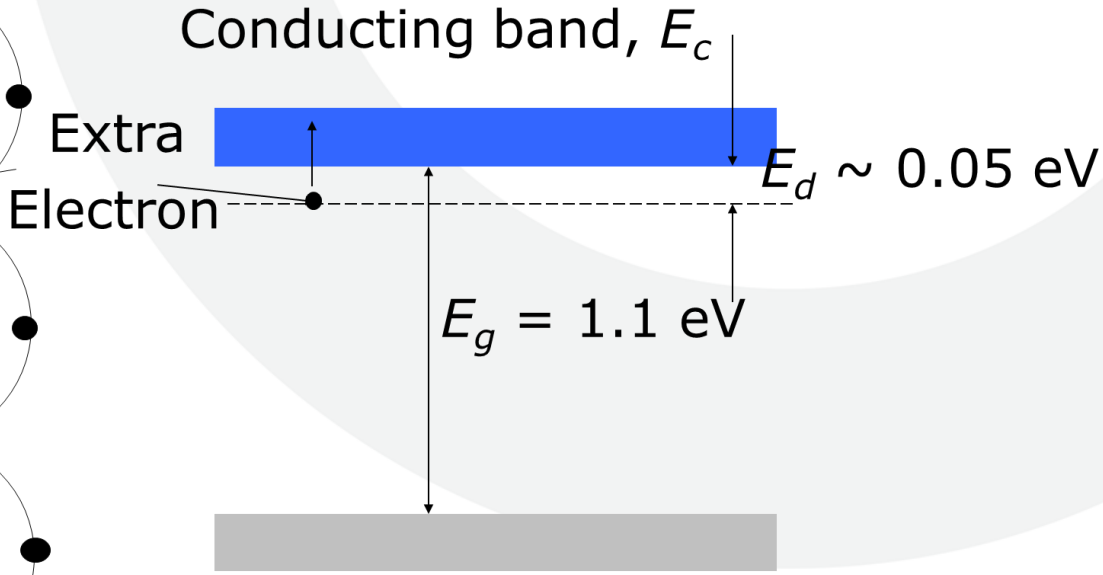




N-type doping

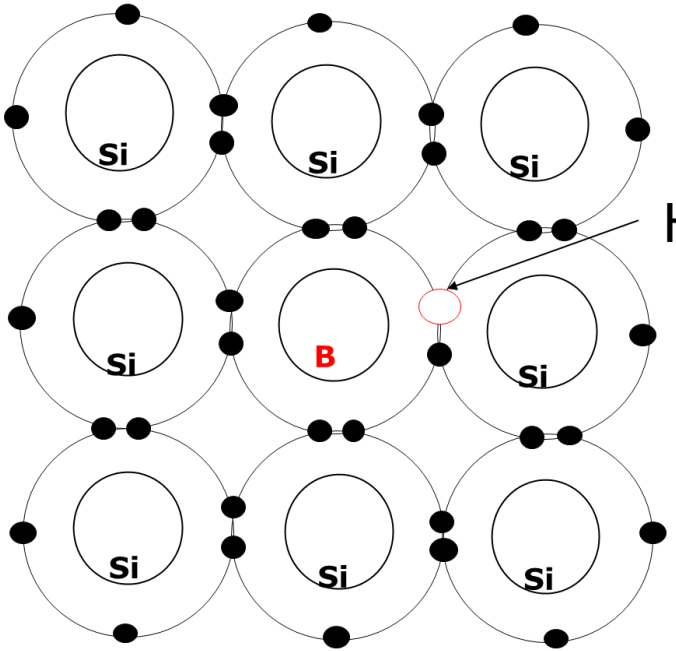
➢ **Doping silicon lattice with group V elements can creates extra electrons in the conduction band — negative charge carriers (n-type), As- donor.**

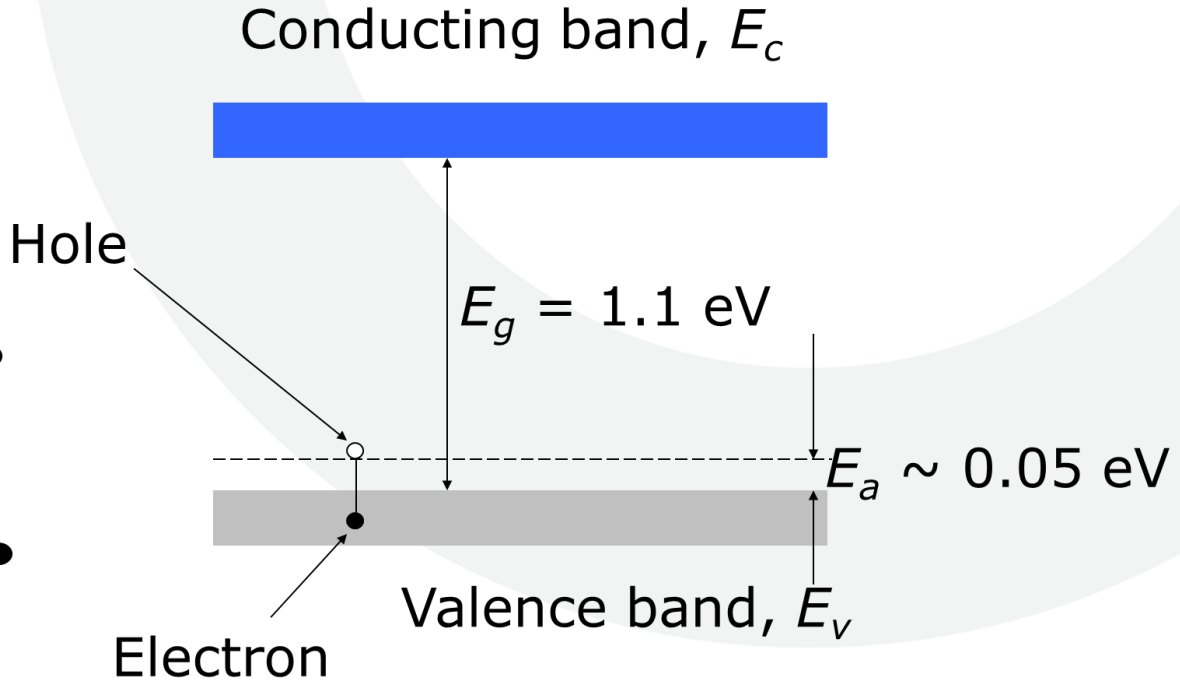




P-type doping

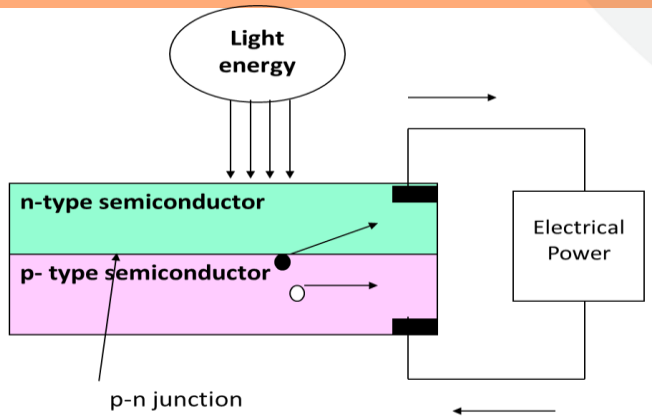
➢ Doping silicon with group III elements can creates empty holes in the conduction band — **positive charge carriers (p-type), B-(acceptor).**

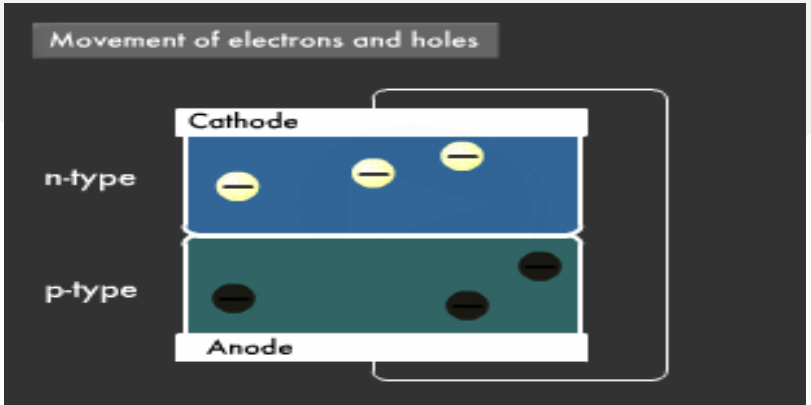




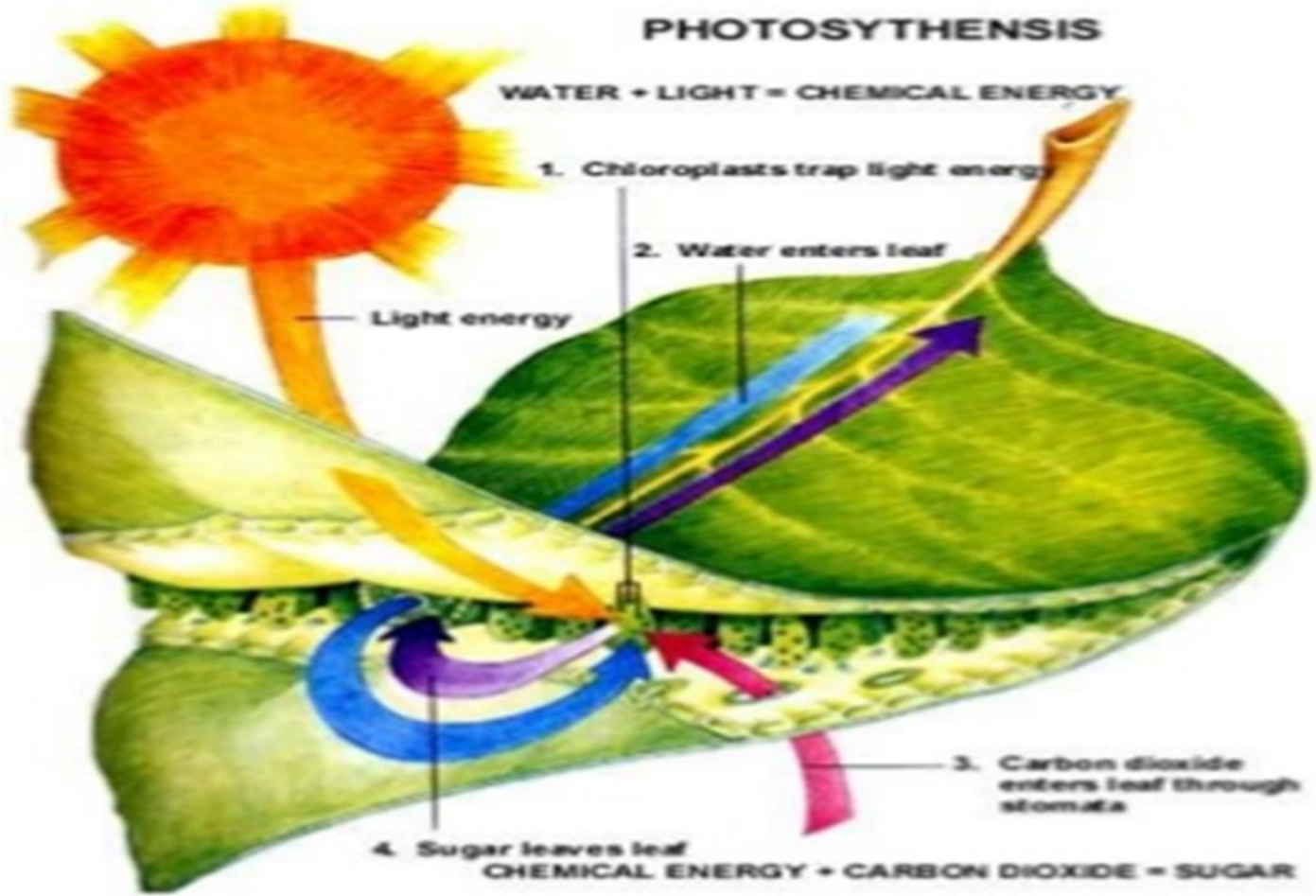
Construction of solar cells

- A typical silicon solar cell consists of n-type material (Phosphorus doped silicon) and a p-type (Boron doped silicon) material made of silicon.
- In the fabrication of device these n-type and p-type materials are joined to form a p-n junction.
- On top of n type connecting front electrodes are placed.
- The metal back contact is attached to p-type layer.
- An antireflective coating is filled in between the front electrodes to avoid the reflection of sunlight.
- Finally, p-type and n-type layers are joined externally to the circuit.
- The complete solar cells are sealed to avoid the environmental contact.

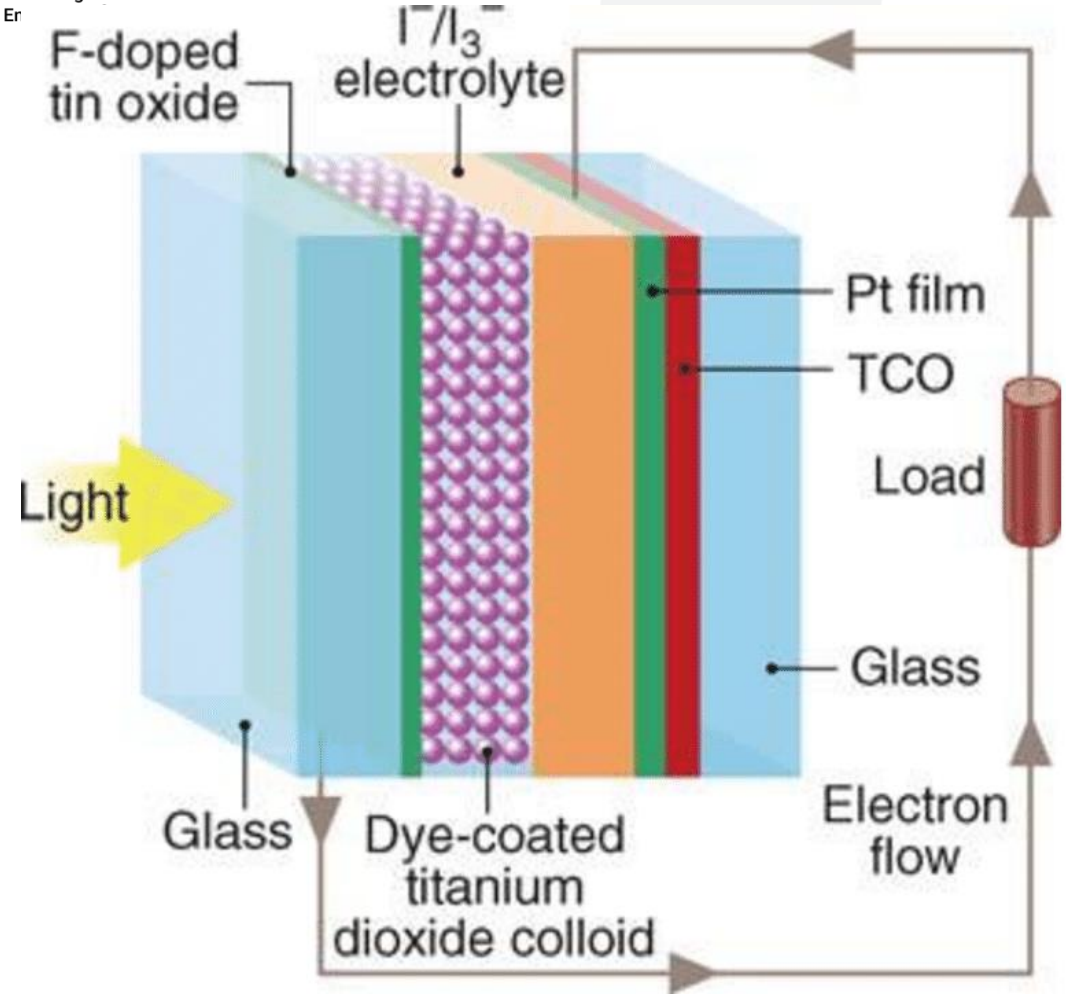




Quantum dot sensitized solar cells



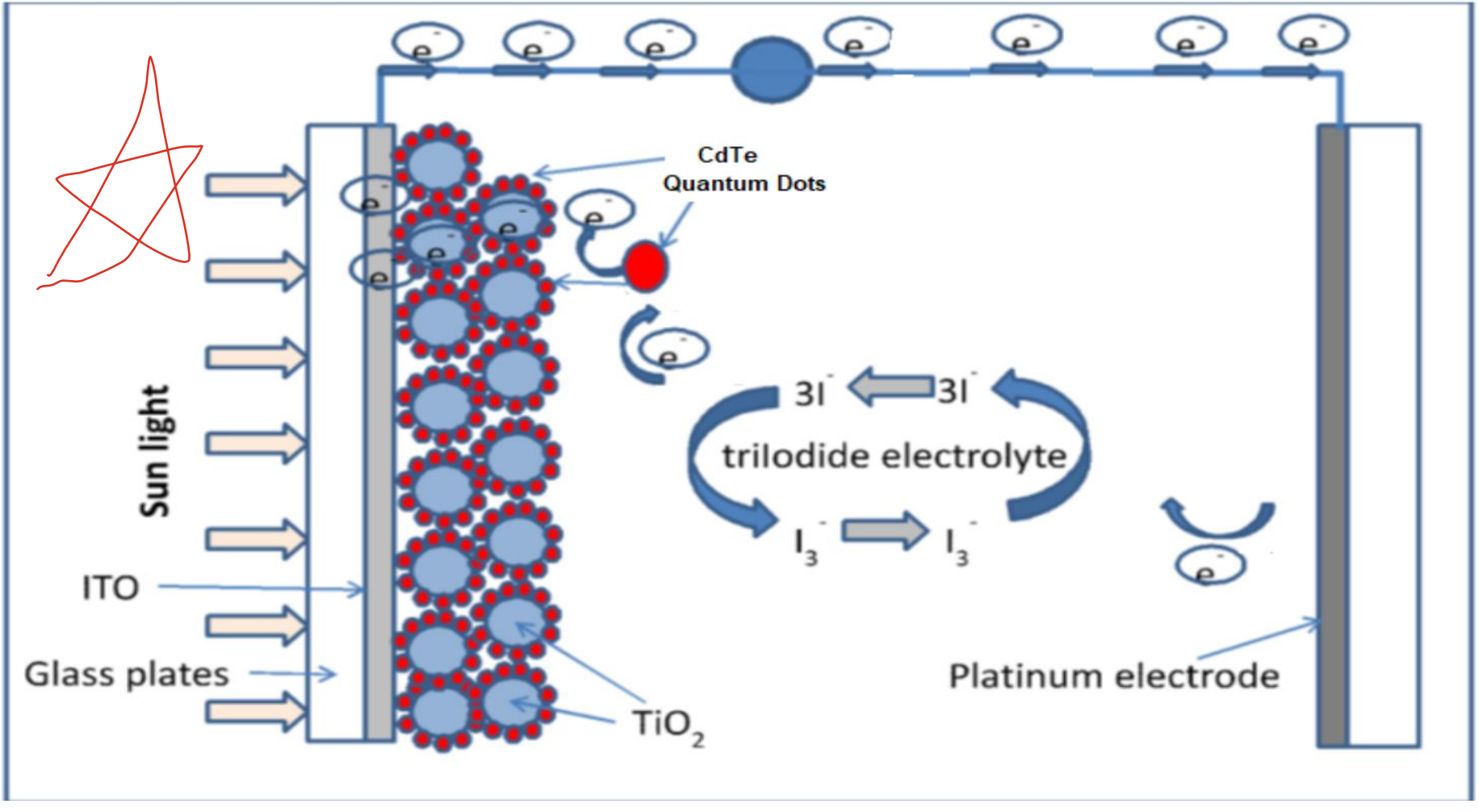
Quantum dot sensitized solar cells





Michael Grätzel

Quantum dot sensitized solar cells



Quantum dot sensitized solar cells

- It consists of two electrodes cathode and anode generally cathode is made of graphite or platinum.
- Anode is made of transparent conducting oxide such as Indium tin oxide. Anode is coated with a layer of nanoporous materials (TiO_2 or ZnO).
- Further these nano porous materials are coated with light absorbing organic/inorganic dye molecules/quantum dots.
- The nano-porous materials acts as photo-sensitizer also it provides road way for electrons.
- A triiodide electrolyte system is employed in between the electrodes, which helps in electron transfer process.

Quantum dot sensitized solar cells

- When light strikes the solar cell, it passes through ITO layer and it reaches the quantum dots.
- Once the photon hits the quantum dot, it undergoes excitation with the liberation of electron, which travels through the network TiO_2 nanoparticles and finally it reaches to cathode.
- The cell is filled with Triiodide electrolyte.
- The electron from the cathode directly reduces the triiodide molecule to iodide molecules.
- Further this iodide molecule reaches the quantum dot oxidizes to form triiodide again with the liberation of electrons, which can be easily absorbed by quantum dots.
- And similar cycle continues whenever sunlight falls on the solar cells.

Quantum dot sensitized solar cells

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